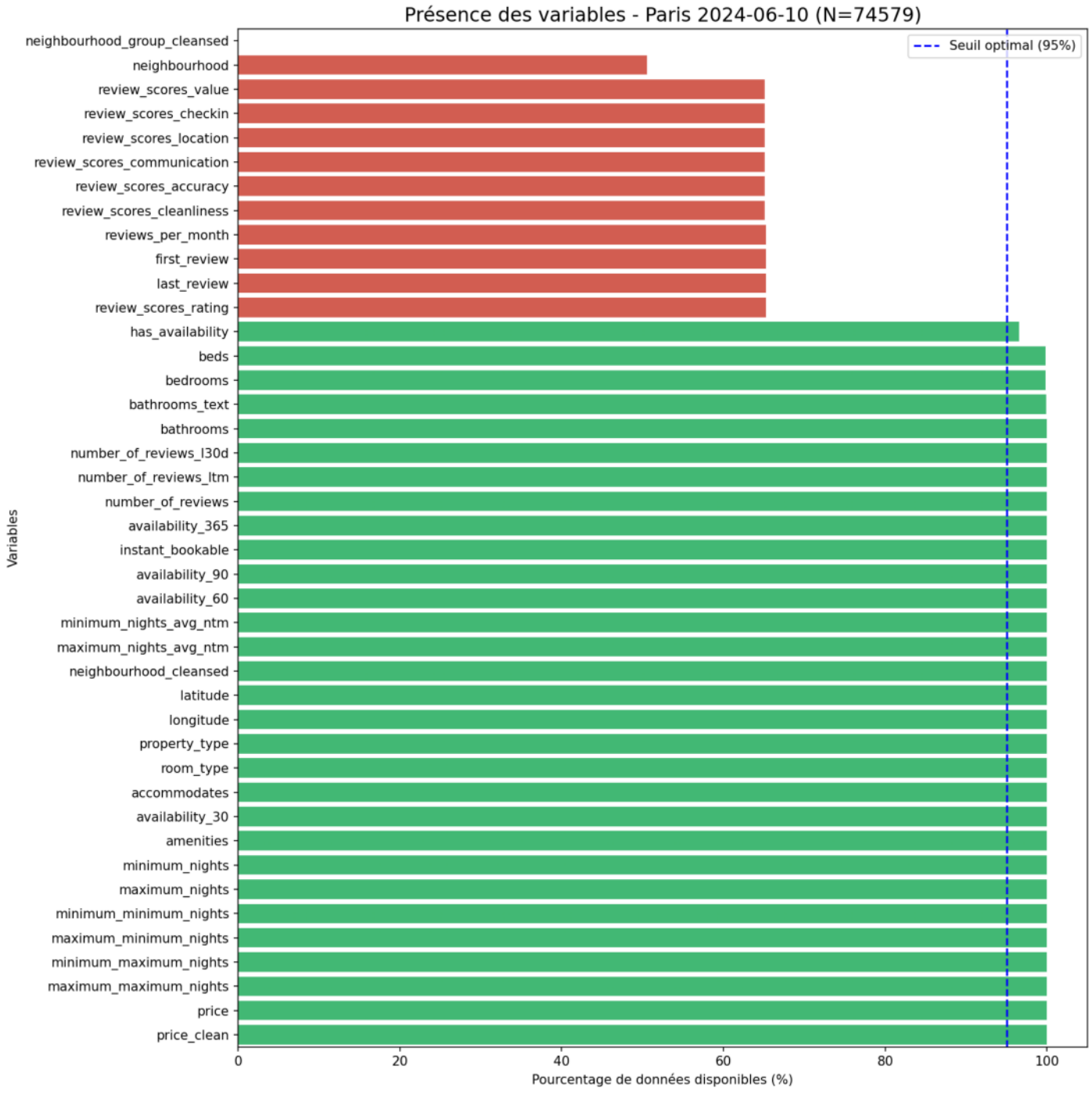
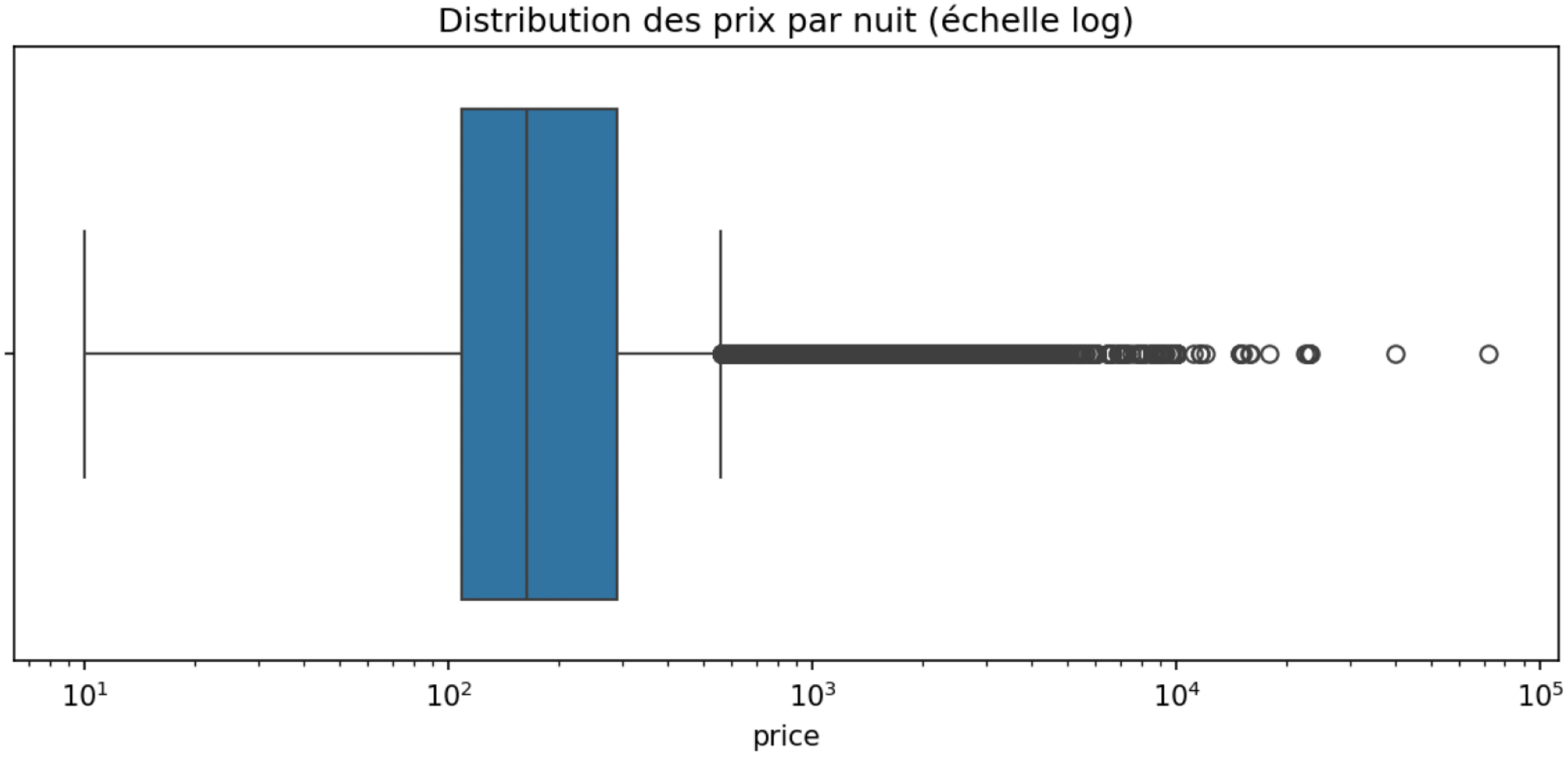
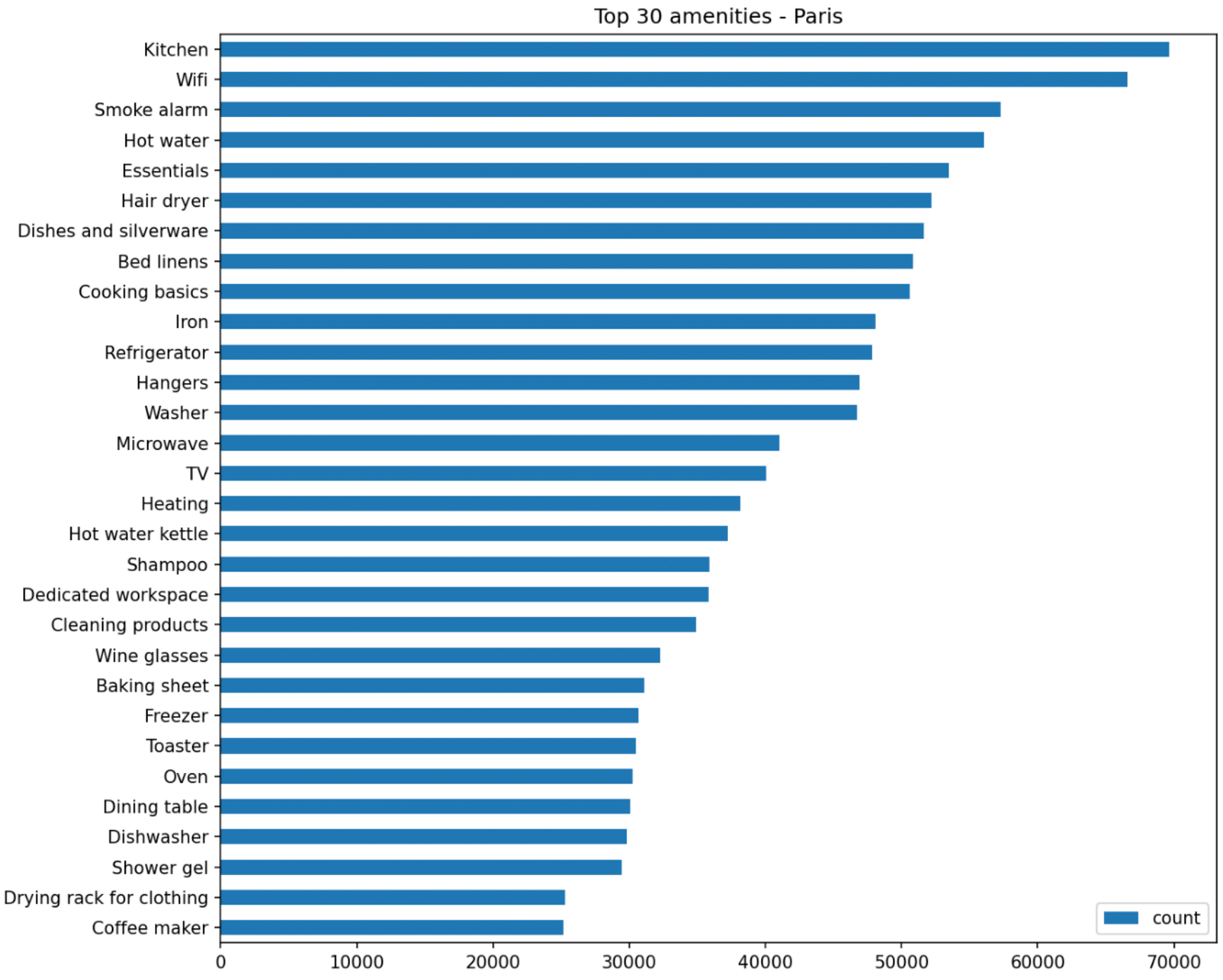
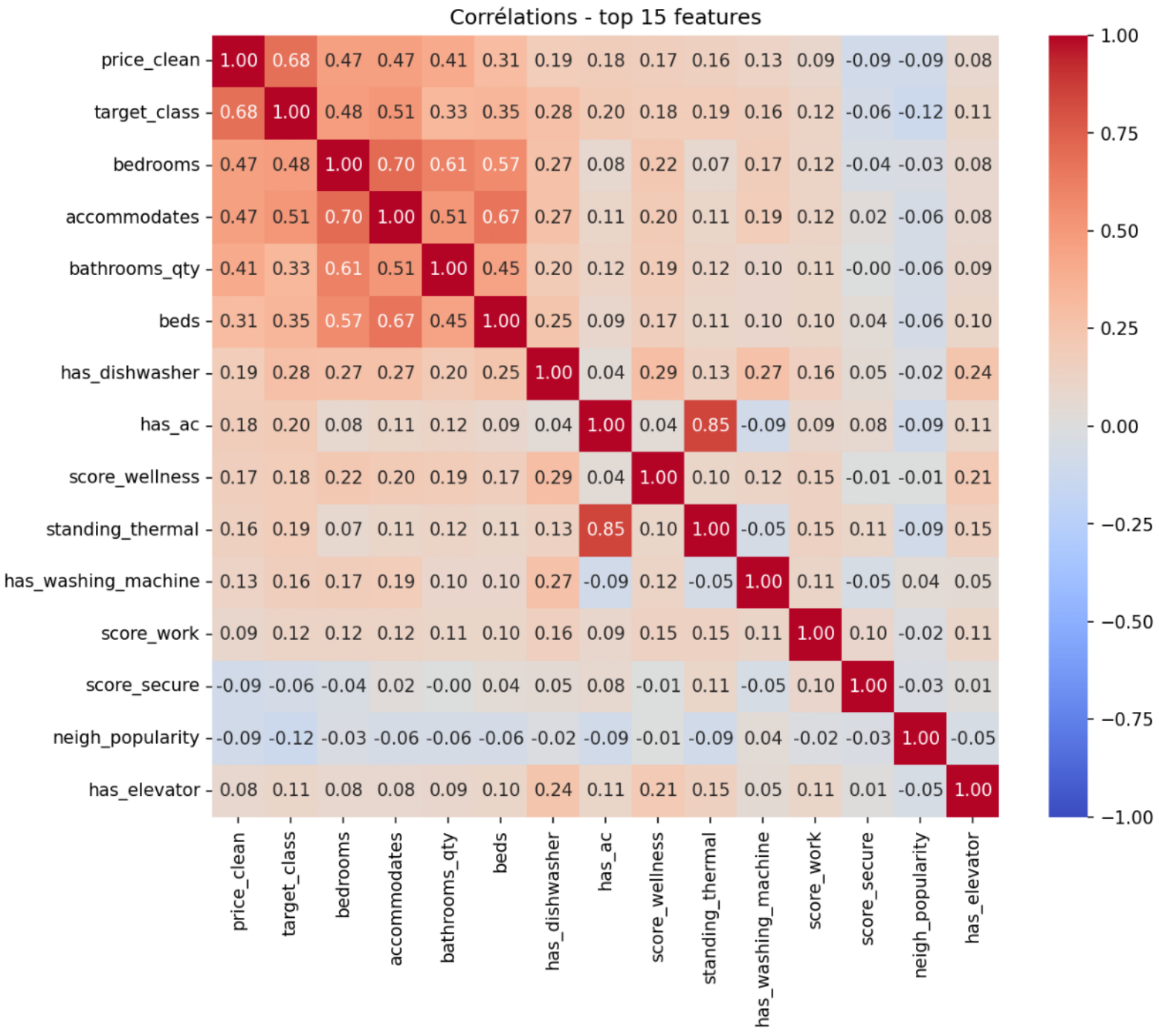
Airbnb price classes — all figures

data_processing: 4 figures
1_ML_Paris: 35 figures
2_generalisation: 6 figures
extensions_pilot: 1 figures

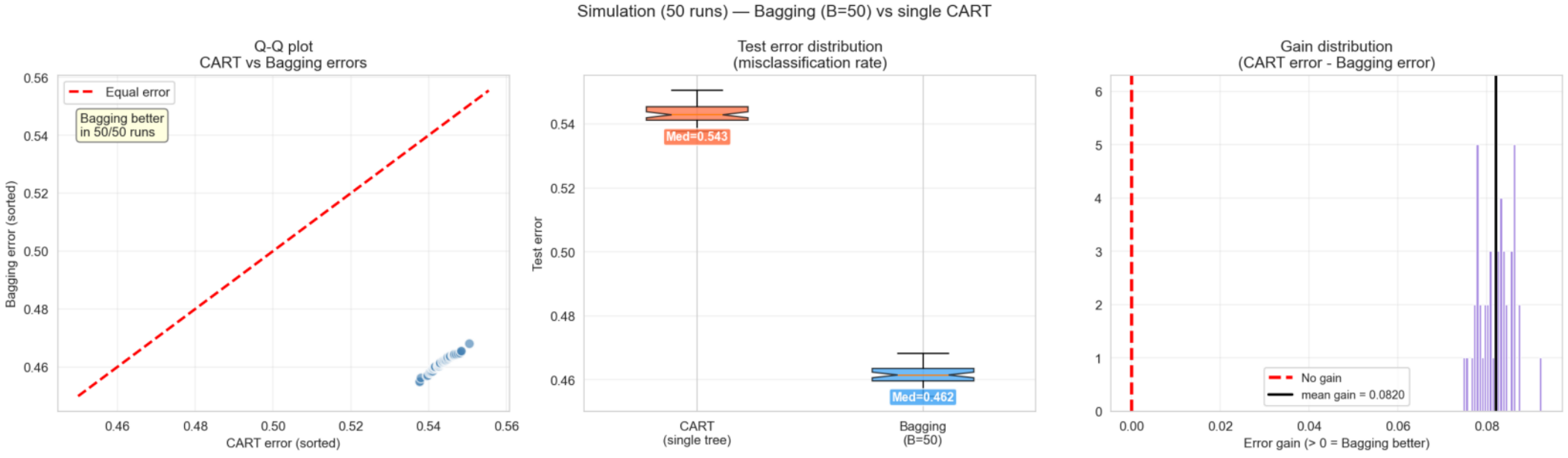

Présence des variables - Paris 2024-06-10 (N=74579)
Seuil optimal (95%)
neighbourhood_group_cleansed
neighbourhood
review_scores_value
review_scores_checkin
review_scores_location
review_scores_communication
review_scores_accuracy
review_scores_cleanliness
reviews_per_month
first_review
last_review
review_scores_rating
has_availability
beds
bedrooms
bathrooms_text
bathrooms
number_of_reviews_l30d
number_of_reviews_ltm
number_of_reviews
availability_365
instant_bookable
availability_90
availability_60
minimum_nights_avg_ntm
maximum_nights_avg_ntm
neighbourhood_cleansed
latitude
longitude
property_type
room_type
accommodates
availability_30
amenities
minimum_nights
maximum_nights
minimum_minimum_nights
maximum_minimum_nights
minimum_maximum_nights
maximum_maximum_nights
price
price_clean
Variables
0
20
40
60
80
100
Pourcentage de données disponibles (%)

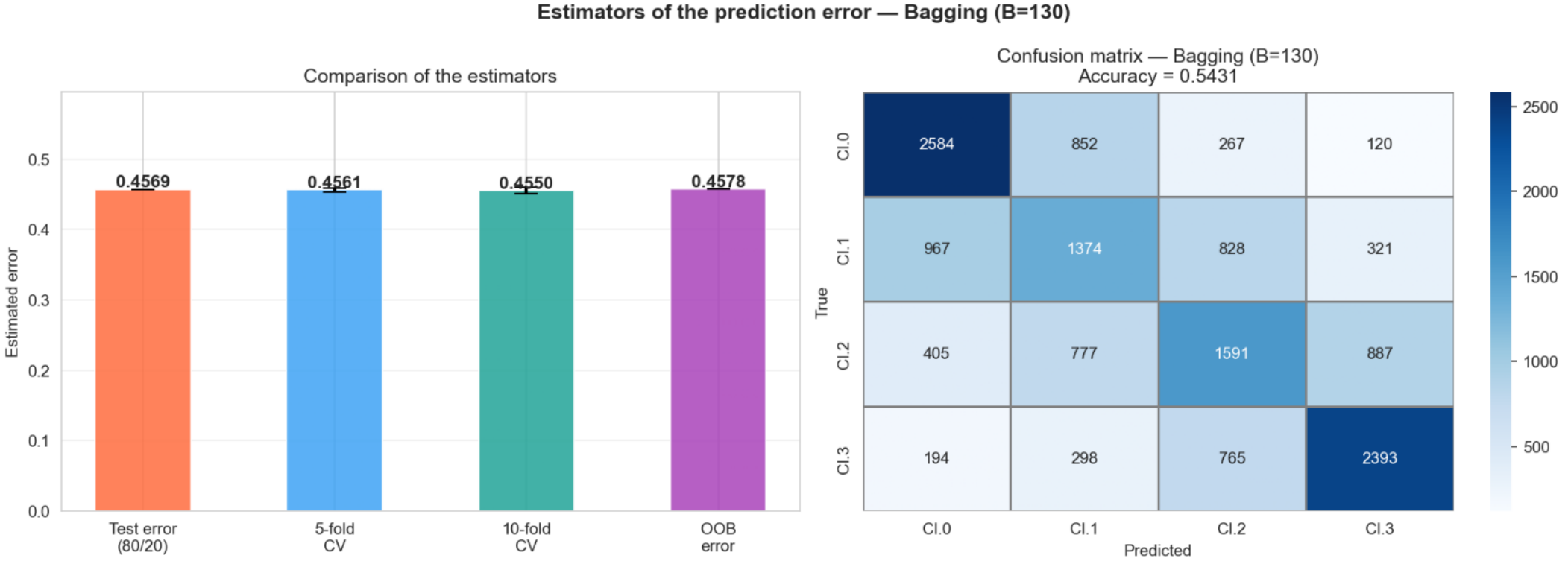

Distribution des prix par nuit (échelle log)
10^1
10^2
10^3
10^4
10^5
price


Top 30 amenities - Paris
Kitchen
Wifi
Smoke alarm
Hot water
Essentials
Hair dryer
Dishes and silverware
Bed linens
Cooking basics
Iron
Refrigerator
Hangers
Washer
Microwave
TV
Heating
Hot water kettle
Shampoo
Dedicated workspace
Cleaning products
Wine glasses
Baking sheet
Freezer
Toaster
Oven
Dining table
Dishwasher
Shower gel
Drying rack for clothing
Coffee maker
0
10000
20000
30000
40000
50000
60000
70000
count

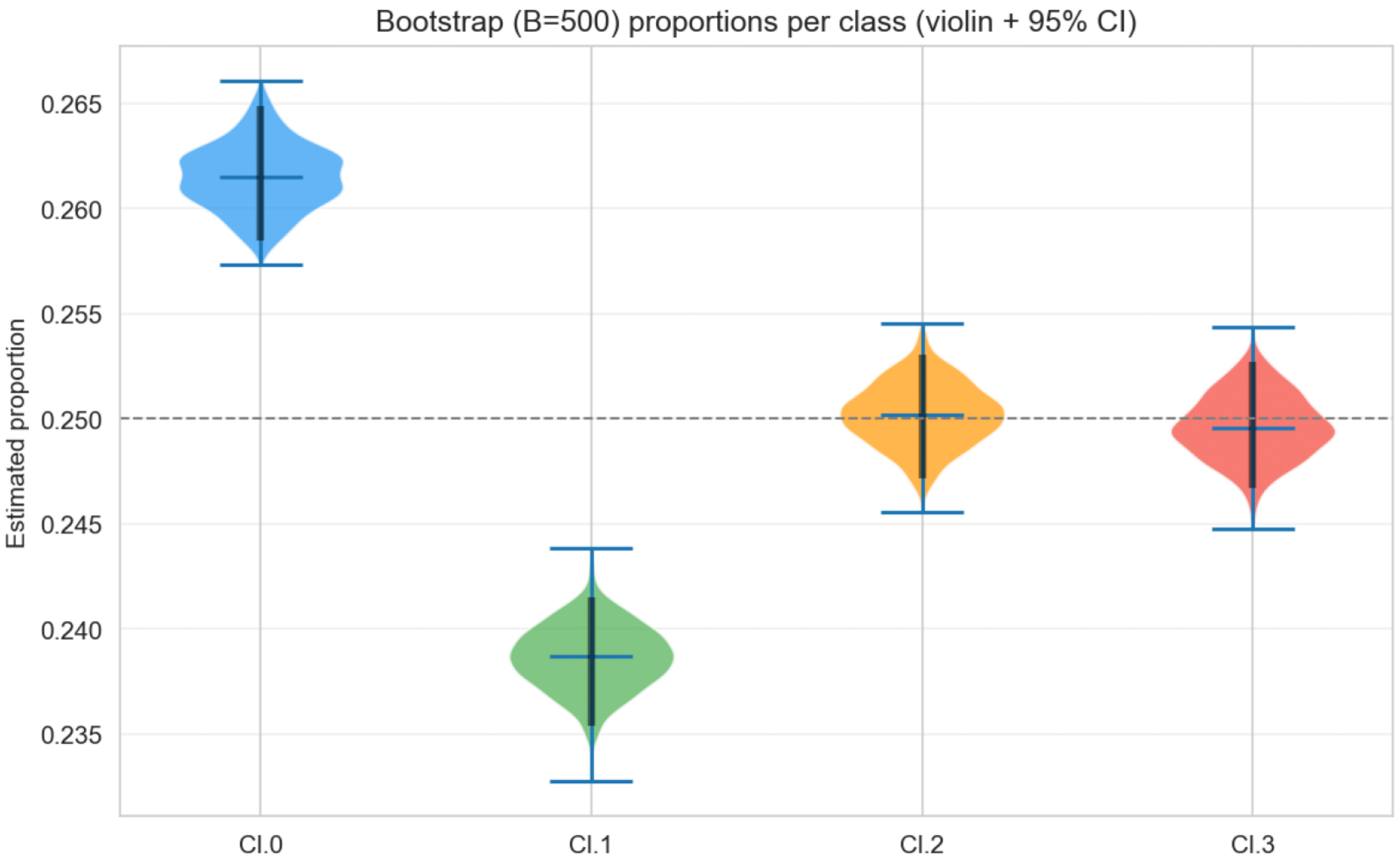

Corrélations - top 15 features
price_clean 1.00 0.68 0.47 0.47 0.41 0.31 0.19 0.18 0.17 0.16 0.13 0.09 -0.09 -0.09 0.08
target_class 0.68 1.00 0.48 0.51 0.33 0.35 0.28 0.20 0.18 0.19 0.16 0.12 -0.06 -0.12 0.11
bedrooms 0.47 0.48 1.00 0.70 0.61 0.57 0.27 0.08 0.22 0.07 0.17 0.12 -0.04 -0.03 0.08
accommodates 0.47 0.51 0.70 1.00 0.51 0.67 0.27 0.11 0.20 0.11 0.19 0.12 0.02 -0.06 0.08
bathrooms_qty 0.41 0.33 0.61 0.51 1.00 0.45 0.20 0.12 0.19 0.12 0.10 0.11 -0.00 -0.06 0.09
beds 0.31 0.35 0.57 0.67 0.45 1.00 0.25 0.09 0.17 0.11 0.10 0.10 0.04 -0.06 0.10
has_dishwasher 0.19 0.28 0.27 0.27 0.20 0.25 1.00 0.04 0.29 0.13 0.27 0.16 0.05 -0.02 0.24
has_ac 0.18 0.20 0.08 0.11 0.12 0.09 0.04 1.00 0.04 0.85 -0.09 0.09 0.08 -0.09 0.11
score_wellness 0.17 0.18 0.22 0.20 0.19 0.17 0.29 0.04 1.00 0.10 0.12 0.15 -0.01 -0.01 0.21
standing_thermal 0.16 0.19 0.07 0.11 0.12 0.11 0.13 0.85 0.10 1.00 -0.05 0.15 0.11 -0.09 0.15
has_washing_machine 0.13 0.16 0.17 0.19 0.10 0.10 0.27 -0.09 0.12 -0.05 1.00 0.11 -0.05 0.04 0.05
score_work 0.09 0.12 0.12 0.12 0.11 0.10 0.16 0.09 0.15 0.15 0.11 1.00 0.10 -0.02 0.11
score_secure -0.09 -0.06 -0.04 0.02 -0.00 0.04 0.05 0.08 -0.01 0.11 -0.05 0.10 1.00 -0.03 0.01
neigh_popularity -0.09 -0.12 -0.03 -0.06 -0.06 -0.06 -0.02 -0.09 -0.01 -0.09 0.04 -0.02 -0.03 1.00 -0.05
has_elevator 0.08 0.11 0.08 0.08 0.09 0.10 0.24 0.11 0.21 0.15 0.05 0.11 0.01 -0.05 1.00
price_clean
target_class
bedrooms
accommodates
bathrooms_qty
beds
has_dishwasher
has_ac
score_wellness
standing_thermal
has_washing_machine
score_work
score_secure
neigh_popularity
has_elevator
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

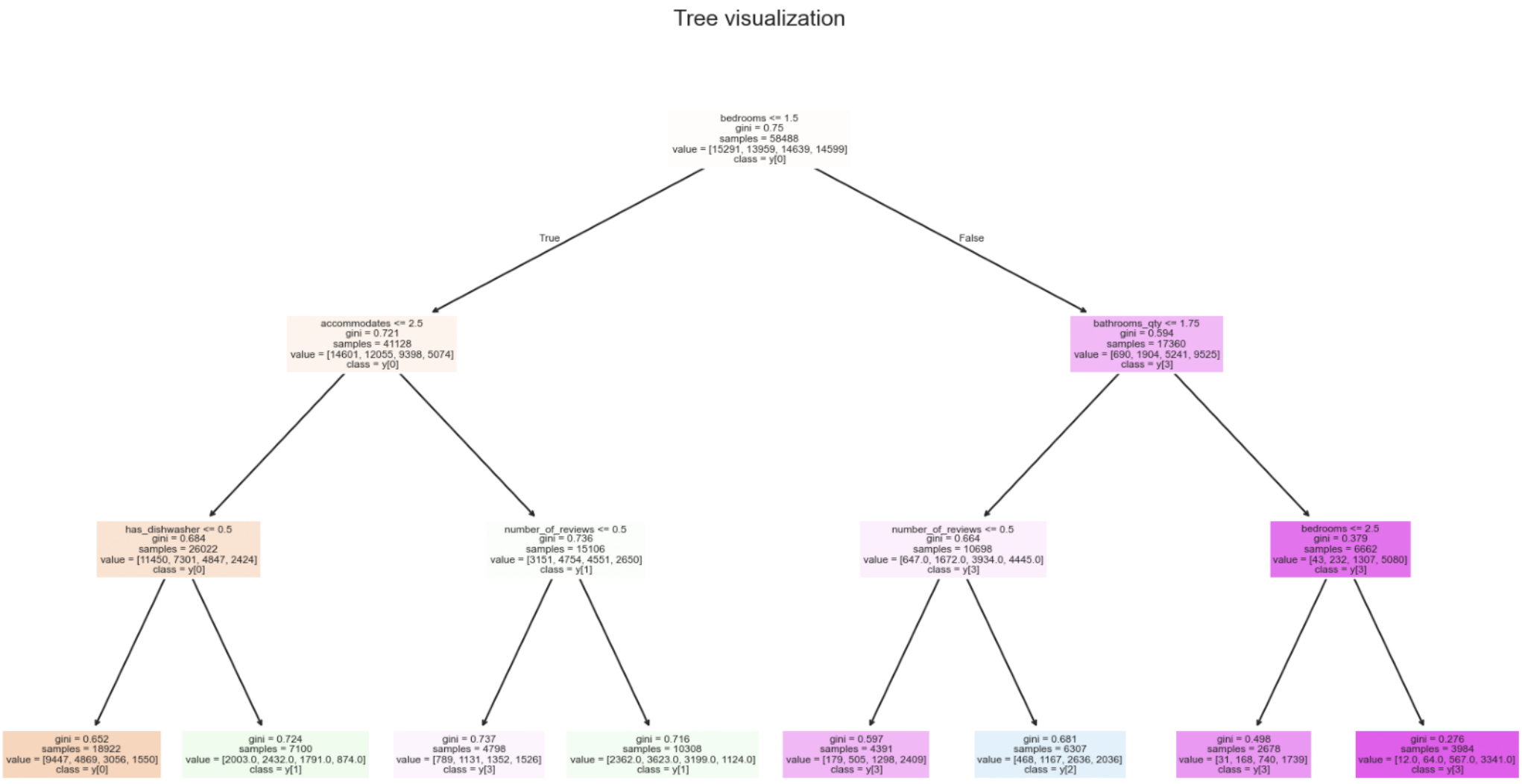

Error vs number of trees B (Bagging of depth-10 trees)
Misclassification rate
0.485
0.480
0.475
0.470
0.465
0.460
0.455
0
50
100
150
200
250
B — number of trees
OOB error (Bagging)
Test error (Bagging)
single CART = 0.4855
best B by OOB = 130
Bagging gain over CART


Simulation (50 runs) — Bagging (B=50) vs single CART
Q-Q plot
CART vs Bagging errors
Equal error
Bagging better
in 50/50 runs
Bagging error (sorted)
CART error (sorted)
0.46
0.48
0.50
0.52
0.54
0.56
Test error distribution
(misclassification rate)
Med=0.543
Med=0.462
Test error
CART
(single tree)
Bagging
(B=50)
Gain distribution
(CART error - Bagging error)
No gain
mean gain = 0.0820
Error gain (> 0 = Bagging better)
0.00
0.02
0.04
0.06
0.08
0
1
2
3
4
5
6

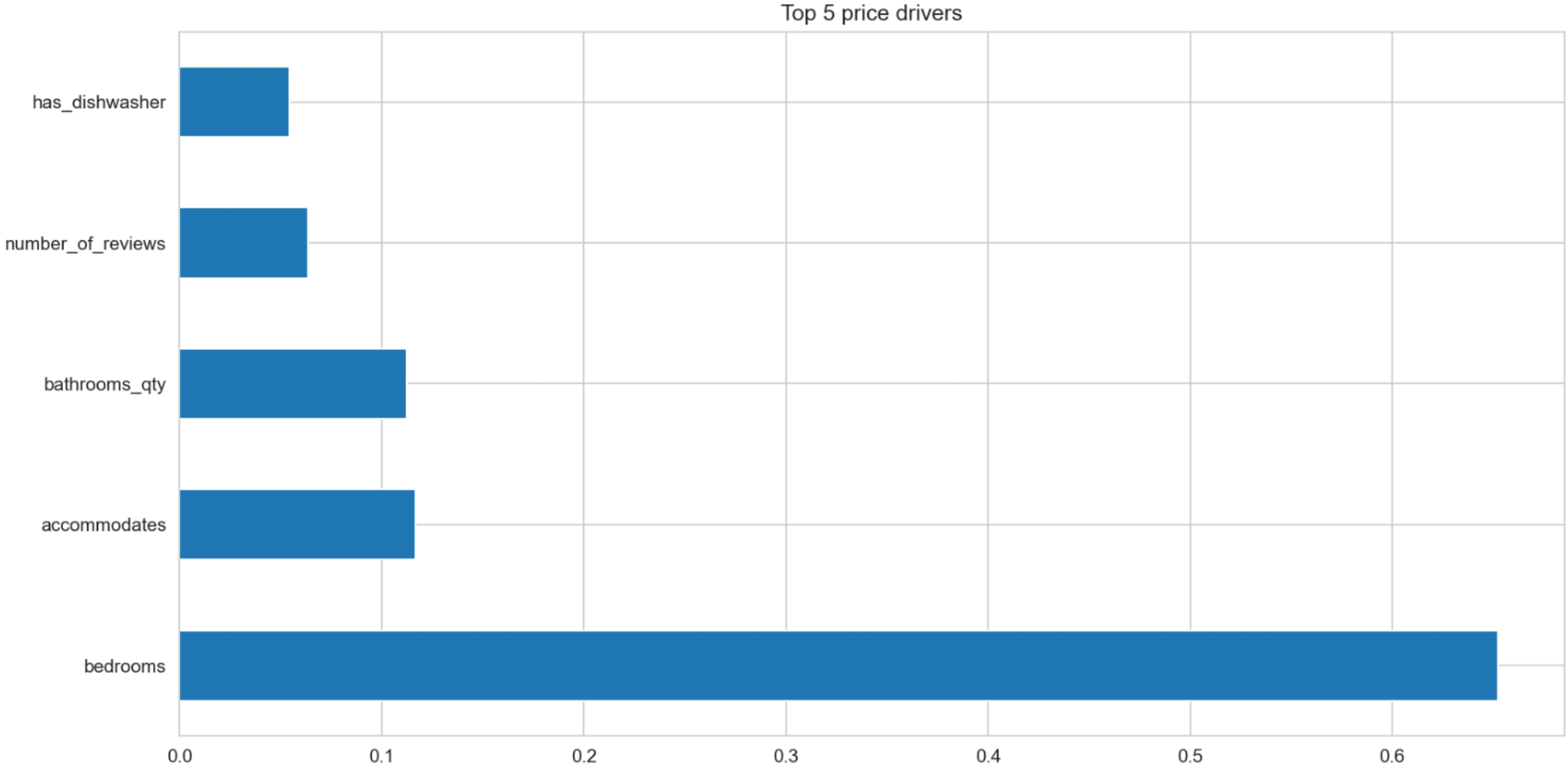

Estimators of the prediction error — Bagging (B=130)
Comparison of the estimators
Estimated error
0.0
0.1
0.2
0.3
0.4
0.5
0.4569
0.4561
0.4550
0.4578
Test error (80/20)
5-fold CV
10-fold CV
OOB error
Confusion matrix — Bagging (B=130)
Accuracy = 0.5431
True
Predicted
Cl.0
Cl.1
Cl.2
Cl.3
2584
852
267
120
967
1374
828
321
405
777
1591
887
194
298
765
2393
2500
2000
1500
1000
500

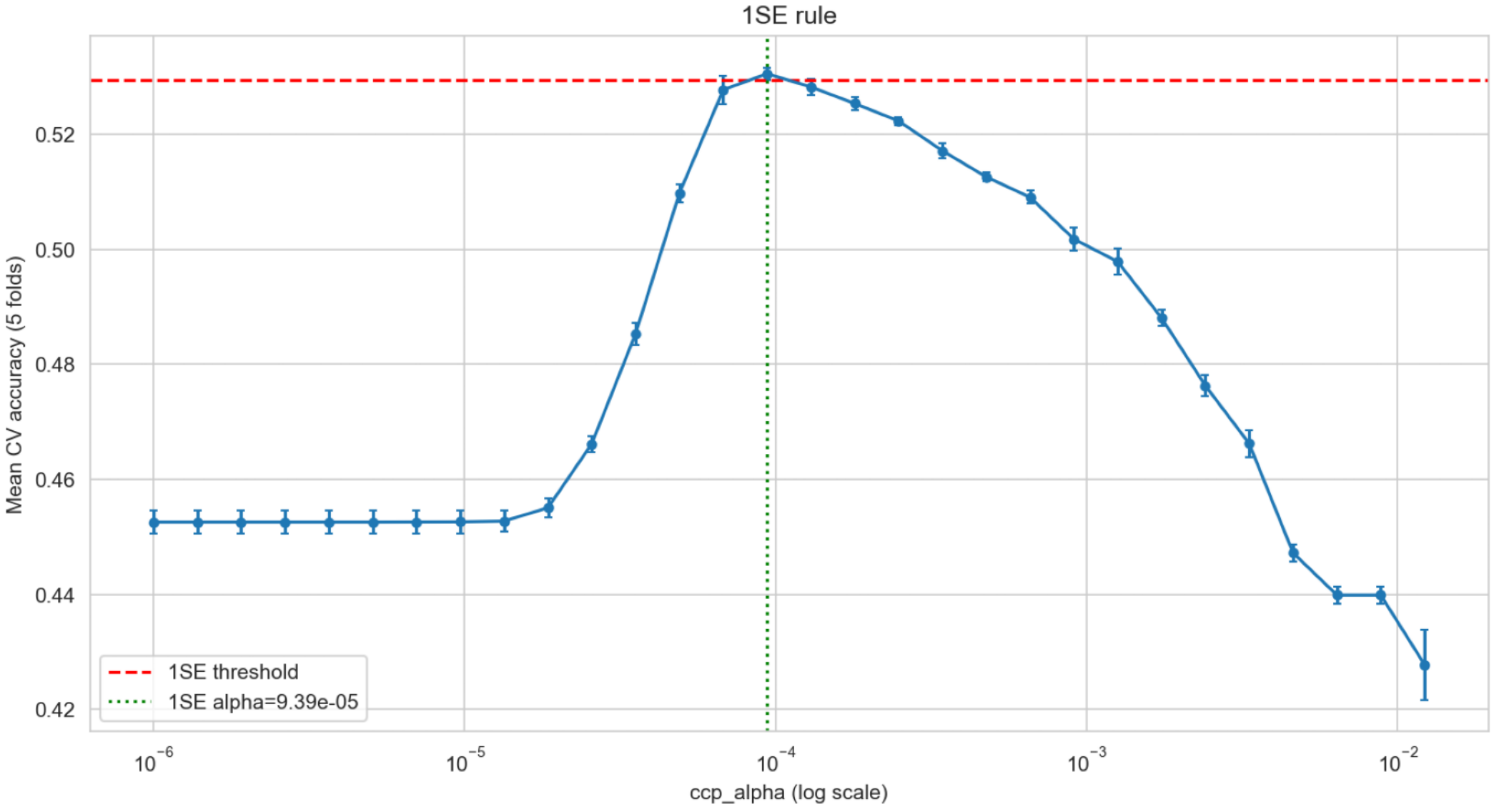

Target variable: target_class (price classes)
Class distribution
Count
19114
(26.1%)
17449
(23.9%)
18299
(25.0%)
18249
(25.0%)
Cl.0 – Very low
Cl.1 – Low-mid
Cl.2 – High-mid
Cl.3 – Very high
Price per night by class (outliers hidden)
Price (log scale)
Cl.0
Cl.1
Cl.2
Cl.3

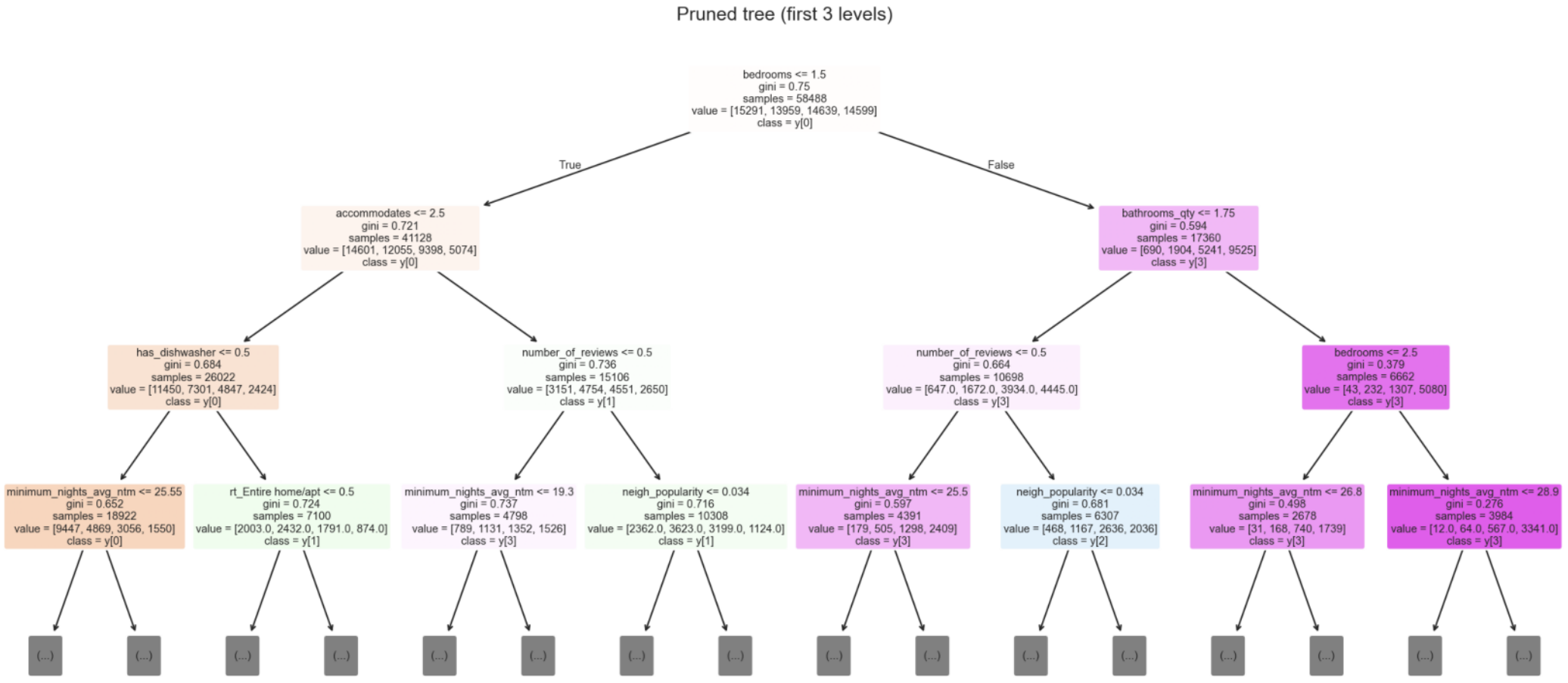

Bootstrap (B=500) proportions per class (violin + 95% CI)
Estimated proportion
0.265
0.260
0.255
0.250
0.245
0.240
0.235
Cl.0
Cl.1
Cl.2
Cl.3

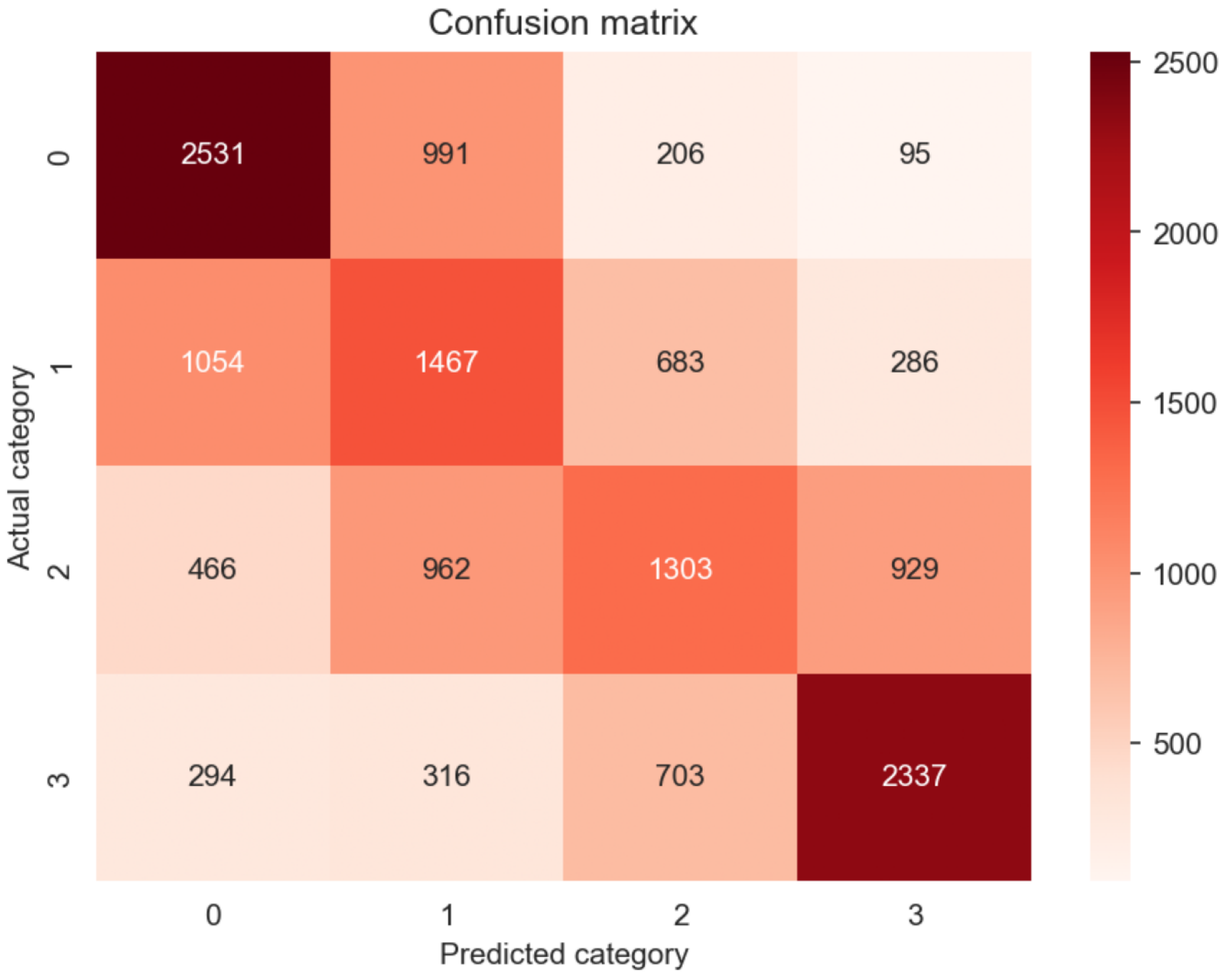

Tree visualization
bedrooms <= 1.5
gini = 0.75
samples = 58488
value = [15291, 13959, 14639, 14599]
class = y[0]
True
False
accommodates <= 2.5
gini = 0.721
samples = 41128
value = [14601, 12055, 9398, 5074]
class = y[0]
bathrooms_qty <= 1.75
gini = 0.594
samples = 17360
value = [690, 1904, 5241, 9525]
class = y[3]
has_dishwasher <= 0.5
gini = 0.684
samples = 26022
value = [11450, 7301, 4847, 2424]
class = y[0]
number_of_reviews <= 0.5
gini = 0.736
samples = 15106
value = [3151, 4754, 4551, 2650]
class = y[1]
number_of_reviews <= 0.5
gini = 0.664
samples = 10698
value = [647.0, 1672.0, 3934.0, 4445.0]
class = y[3]
bedrooms <= 2.5
gini = 0.379
samples = 6662
value = [43, 232, 1307, 5080]
class = y[3]
gini = 0.652
samples = 18922
value = [9447, 4869, 3056, 1550]
class = y[0]
gini = 0.724
samples = 7100
value = [2003.0, 2432.0, 1791.0, 874.0]
class = y[1]
gini = 0.737
samples = 4798
value = [789, 1131, 1352, 526]
class = y[3]
gini = 0.716
samples = 10308
value = [2362.0, 3623.0, 3199.0, 1124.0]
class = y[1]
gini = 0.597
samples = 4391
value = [179, 505, 1298, 2409]
class = y[3]
gini = 0.681
samples = 6307
value = [468, 1167, 2636, 2036]
class = y[2]
gini = 0.498
samples = 2678
value = [31, 168, 740, 1739]
class = y[3]
gini = 0.276
samples = 3984
value = [12.0, 64.0, 567.0, 3341.0]
class = y[3]

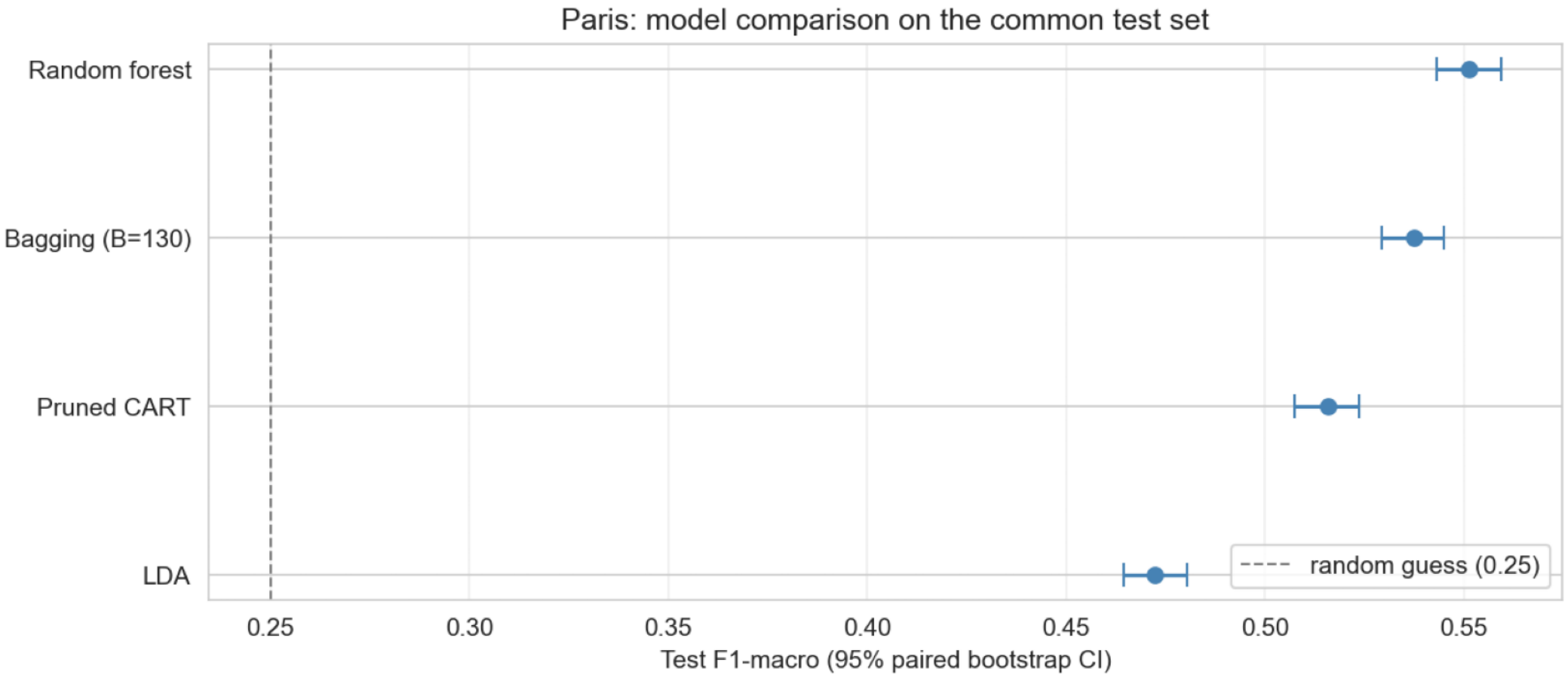
1_ML_Paris / cart_02_depth3_top5_importance

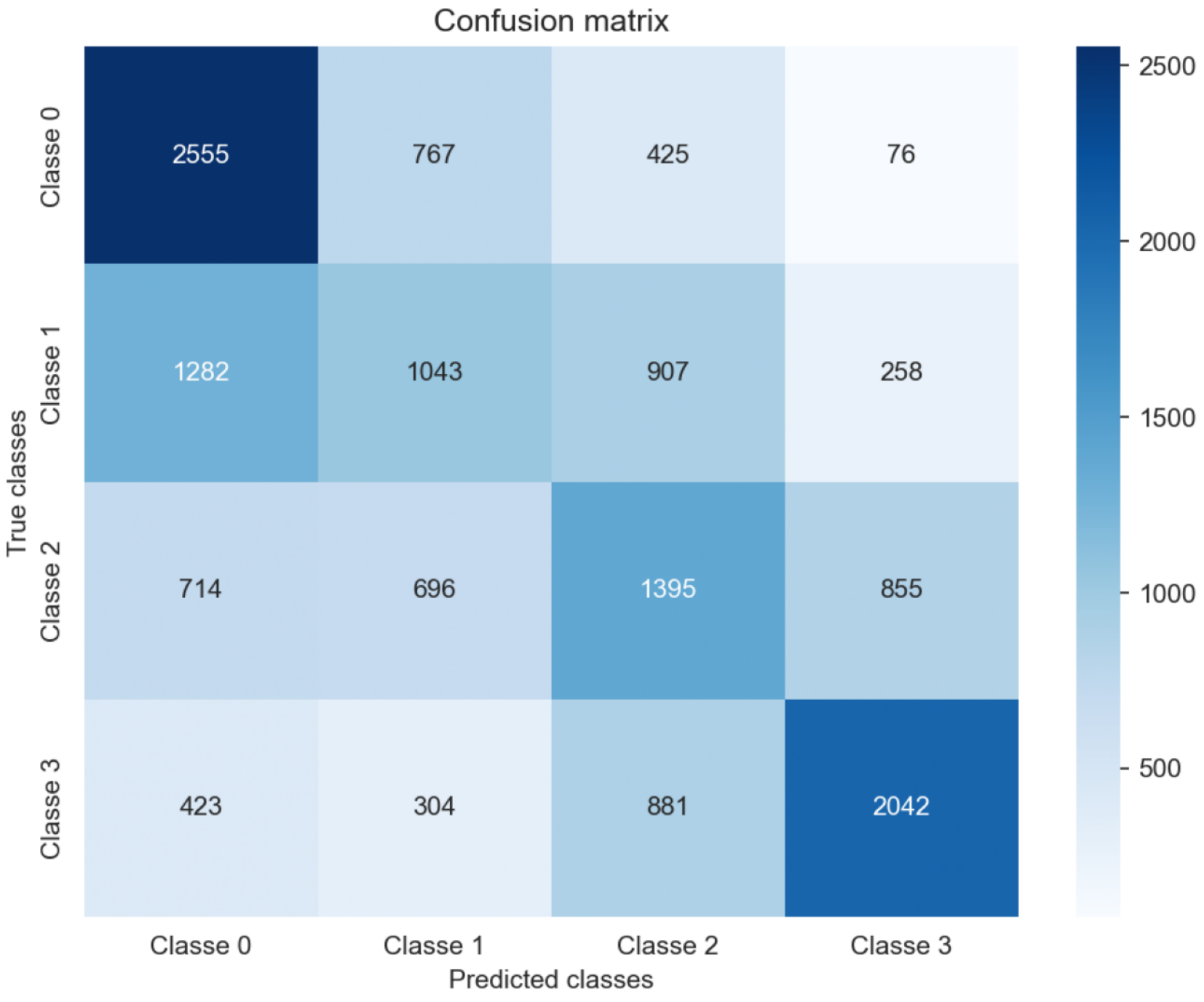


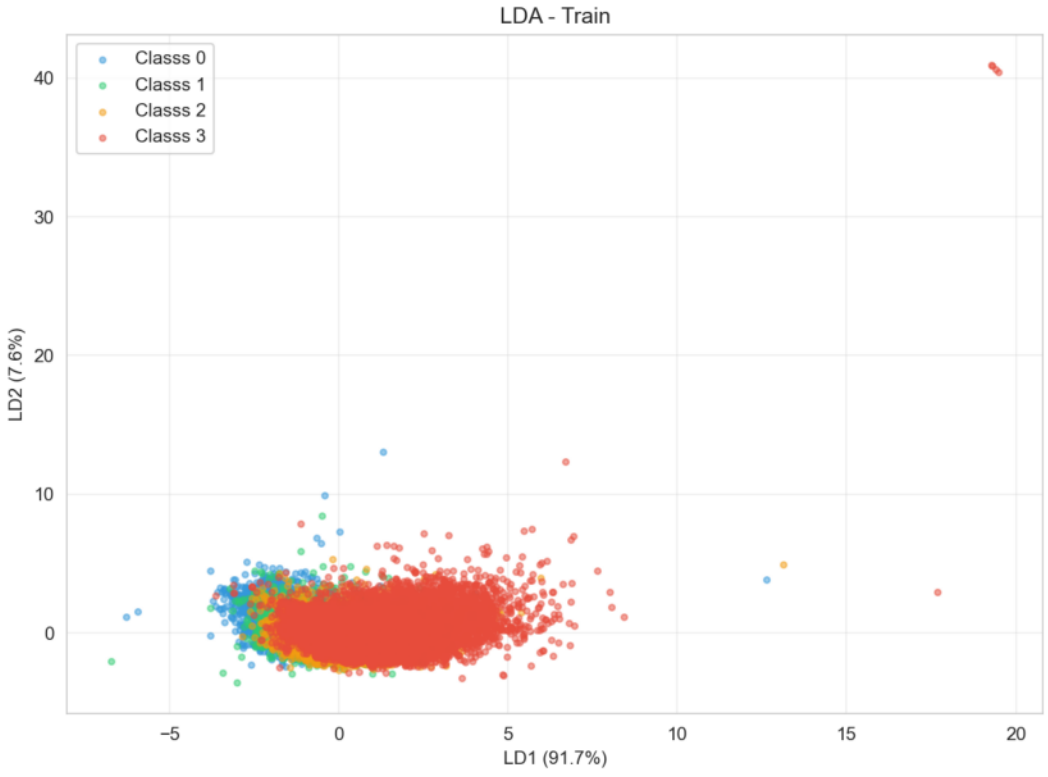

1SE rule
Mean CV accuracy (5 folds)
0.52
0.50
0.48
0.46
0.44
0.42
1SE threshold
1SE alpha=9.39e-05
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
ccp_alpha (log scale)

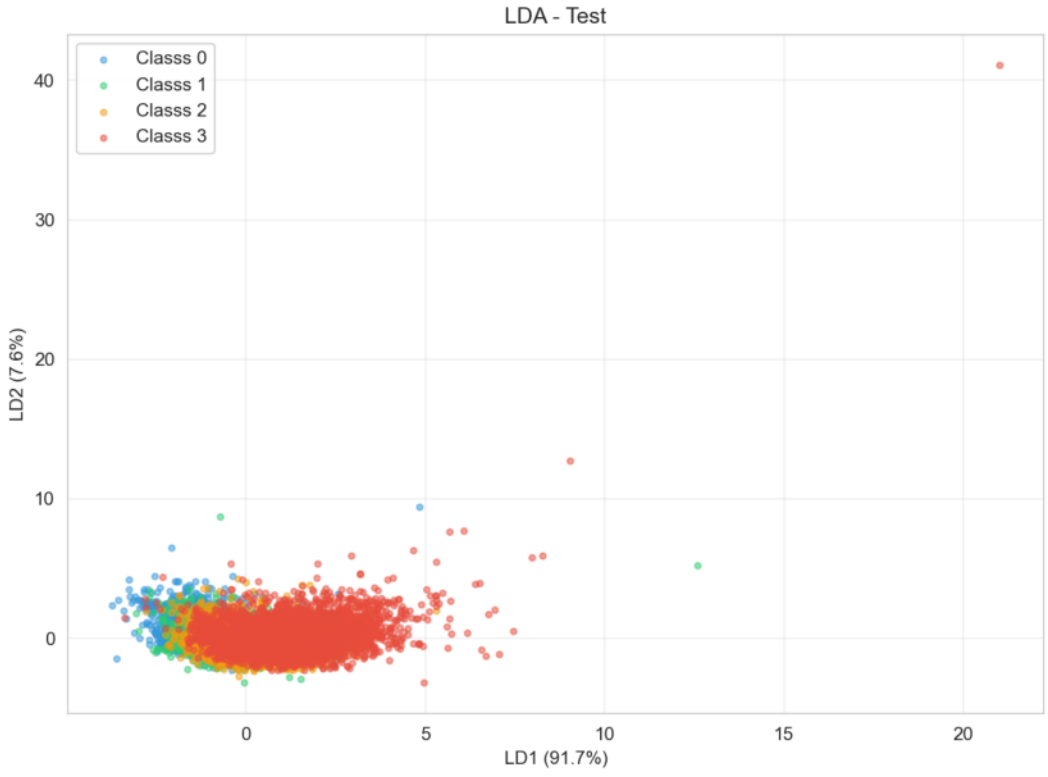

Pruned tree (first 3 levels)
bedrooms <= 1.5
gini = 0.75
samples = 58488
value = [15291, 13959, 14639, 14599]
class = y[0]
True
False
accommodates <= 2.5
gini = 0.721
samples = 41128
value = [14601, 12055, 9398, 5074]
class = y[0]
bathrooms_qty <= 1.75
gini = 0.594
samples = 17360
value = [690, 1904, 5241, 9525]
class = y[3]
has_dishwasher <= 0.5
gini = 0.684
samples = 26022
value = [11450, 7301, 4847, 2424]
class = y[0]
number_of_reviews <= 0.5
gini = 0.736
samples = 15106
value = [3151, 4754, 4551, 2650]
class = y[1]
number_of_reviews <= 0.5
gini = 0.664
samples = 10698
value = [647.0, 1672.0, 3934.0, 4445.0]
class = y[3]
bedrooms <= 2.5
gini = 0.379
samples = 6662
value = [43, 232, 1307, 5080]
class = y[3]
minimum_nights_avg_ntm <= 25.55
gini = 0.652
samples = 18922
value = [9447, 4869, 3056, 1550]
class = y[0]
rt_Entire home/apt <= 0.5
gini = 0.724
samples = 7100
value = [2003.0, 2432.0, 1791.0, 874.0]
class = y[1]
minimum_nights_avg_ntm <= 19.3
gini = 0.737
samples = 4798
value = [789, 1131, 1352, 1526]
class = y[3]
neigh_popularity <= 0.034
gini = 0.716
samples = 10308
value = [2362.0, 3623.0, 3199.0, 1124.0]
class = y[1]
minimum_nights_avg_ntm <= 25.5
gini = 0.597
samples = 4391
value = [179, 505, 1298, 2409]
class = y[3]
neigh_popularity <= 0.034
gini = 0.681
samples = 6307
value = [468, 1167, 2636, 2036]
class = y[2]
minimum_nights_avg_ntm <= 26.8
gini = 0.498
samples = 2678
value = [31, 168, 740, 1739]
class = y[3]
minimum_nights_avg_ntm <= 28.9
gini = 0.276
samples = 3984
value = [12.0, 64.0, 567.0, 3341.0]
class = y[3]
(...)

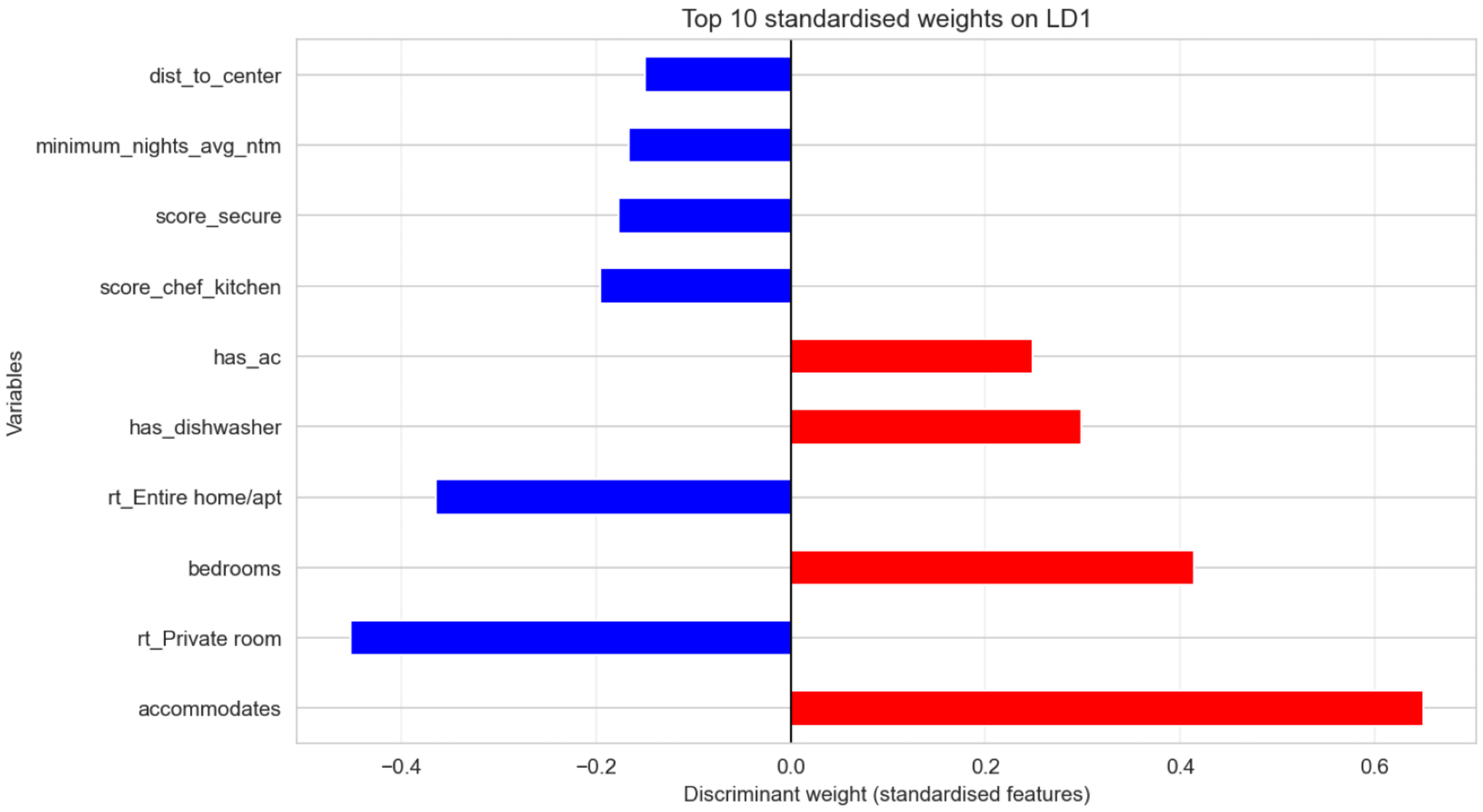

Confusion matrix
2531
991
206
95
1054
1467
683
286
466
962
1303
929
294
316
703
2337
Actual category
0
1
2
3
Predicted category
0
1
2
3
2500
2000
1500
1000
500

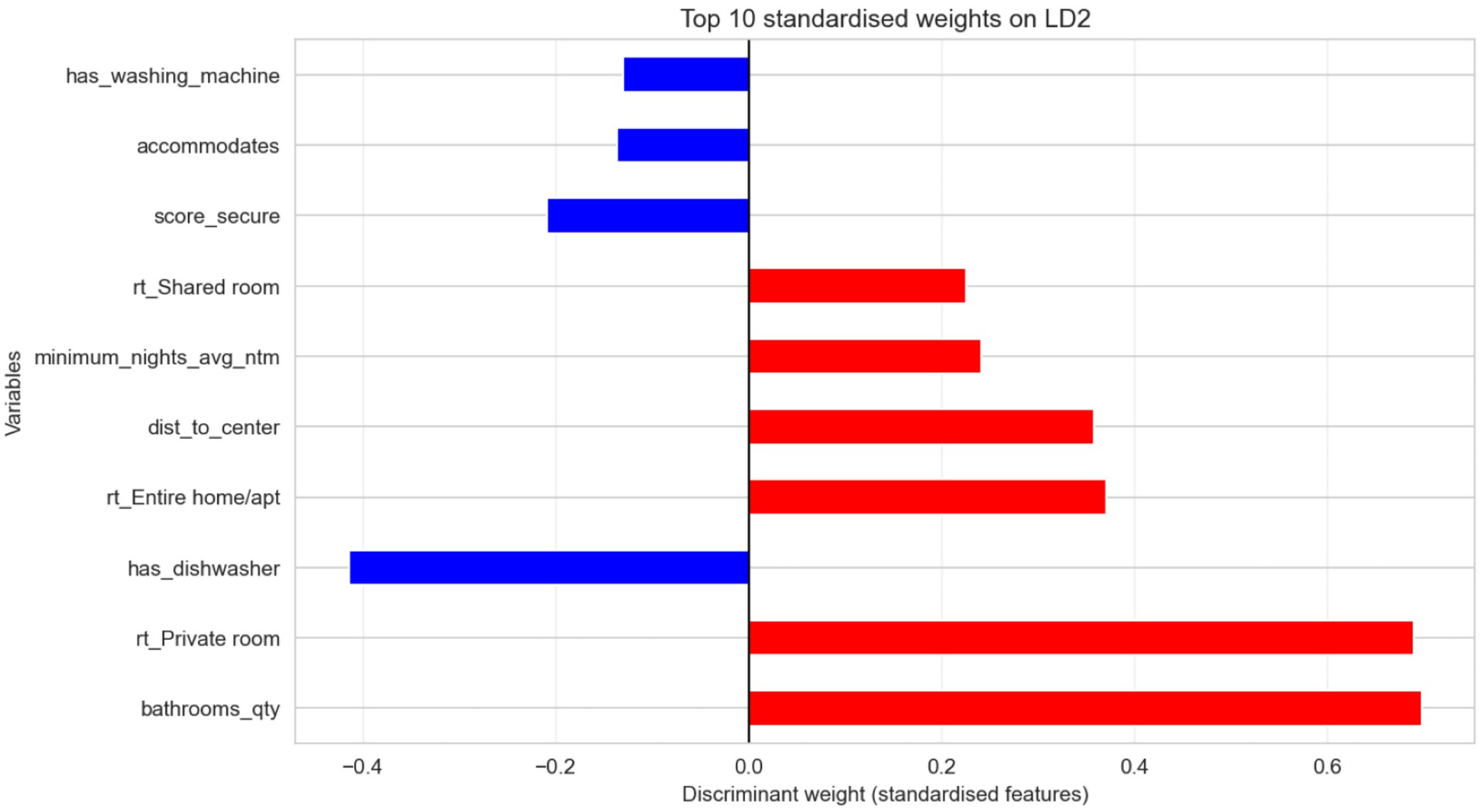

Paris: model comparison on the common test set
Random forest
Bagging (B=130)
Pruned CART
LDA
random guess (0.25)
0.25
0.30
0.35
0.40
0.45
0.50
0.55
Test F1-macro (95% paired bootstrap CI)

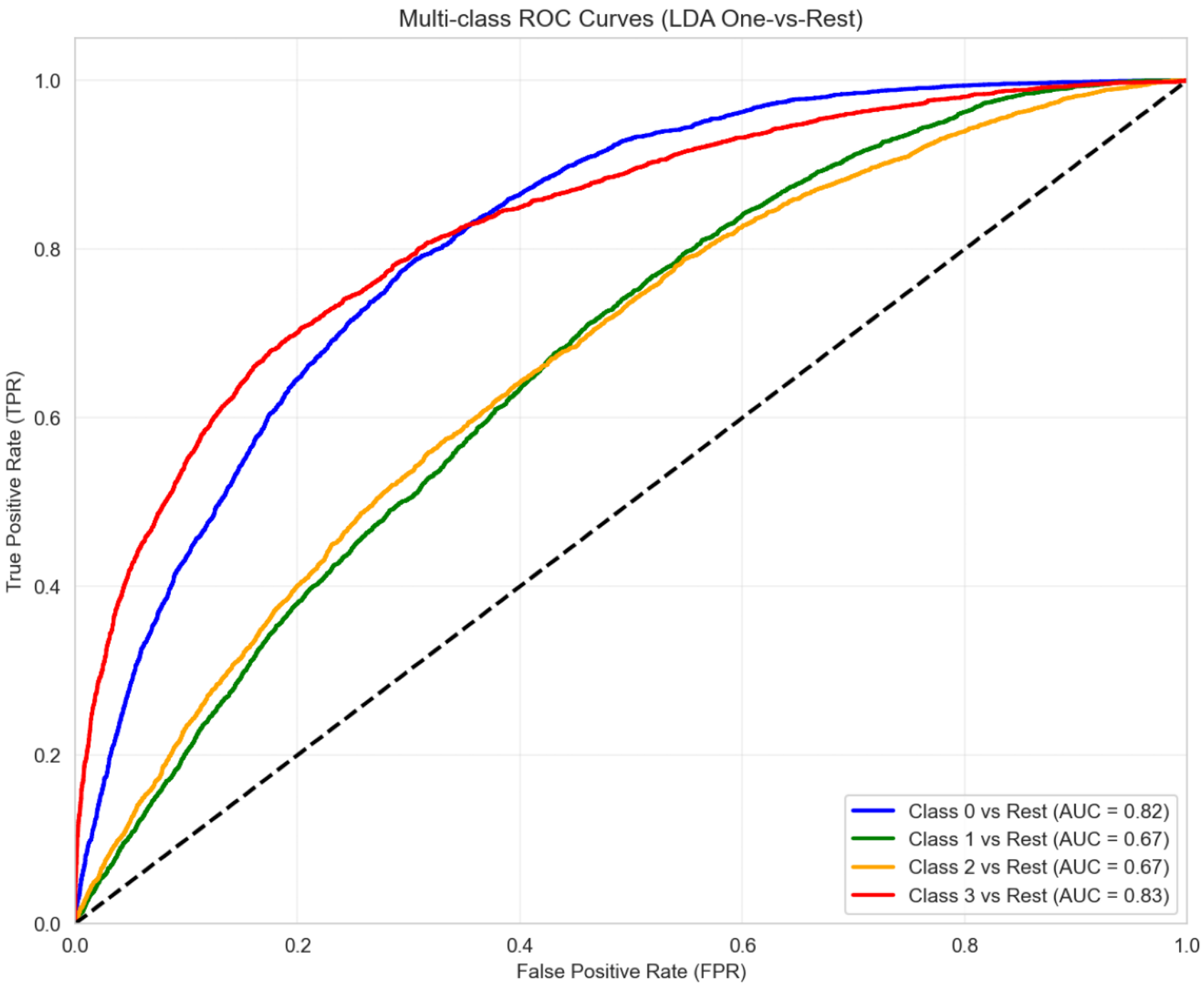

Confusion matrix
2555
767
425
76
1282
1043
907
258
714
696
1395
855
423
304
881
2042
Classe 0
Classe 1
Classe 2
Classe 3
True classes
Classe 0
Classe 1
Classe 2
Classe 3
Predicted classes
2500
2000
1500
1000
500

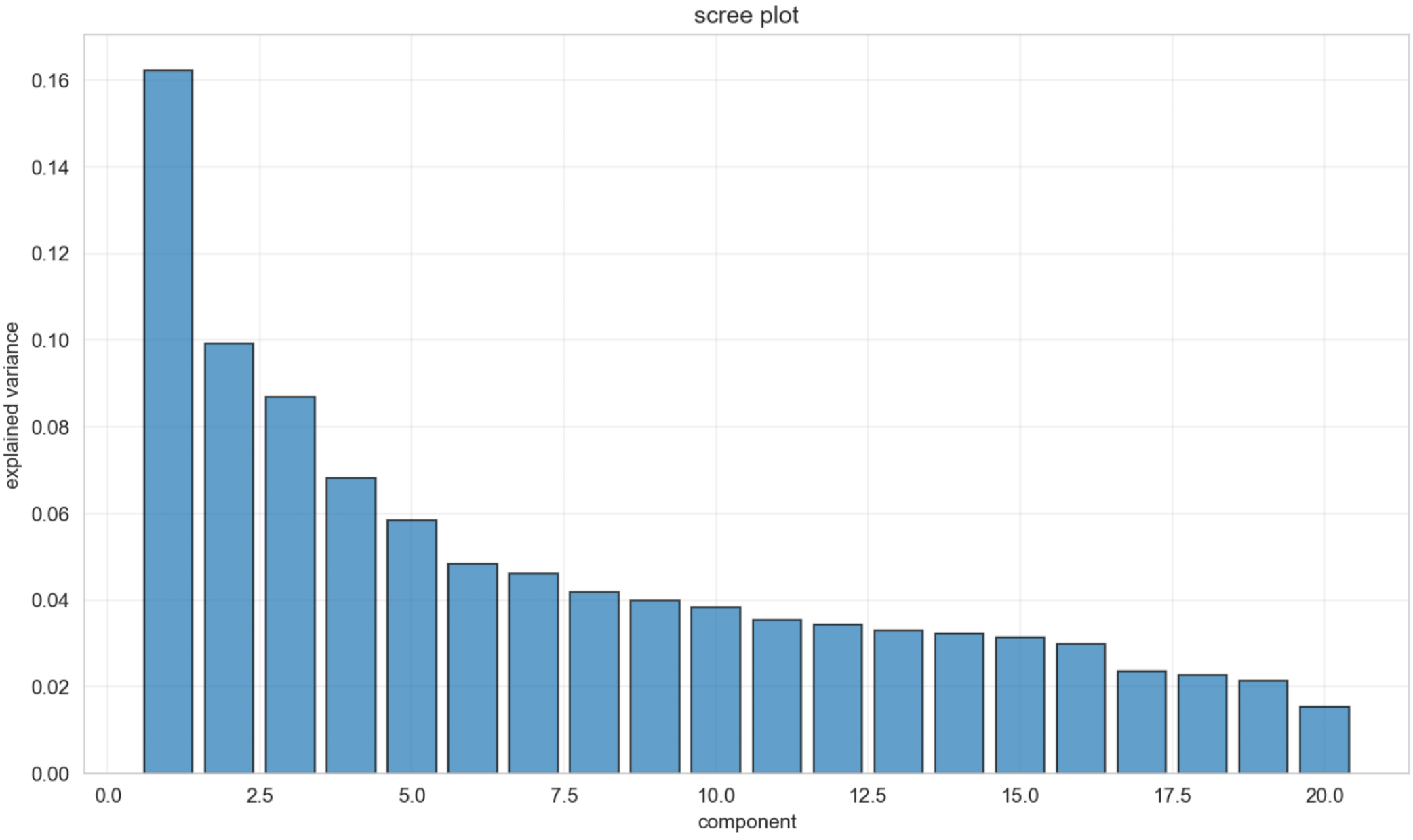

LDA - Train
Classs 0
Classs 1
Classs 2
Classs 3
LD2 (7.6%)
LD1 (91.7%)
40
30
20
10
0
−5
0
5
10
15
20

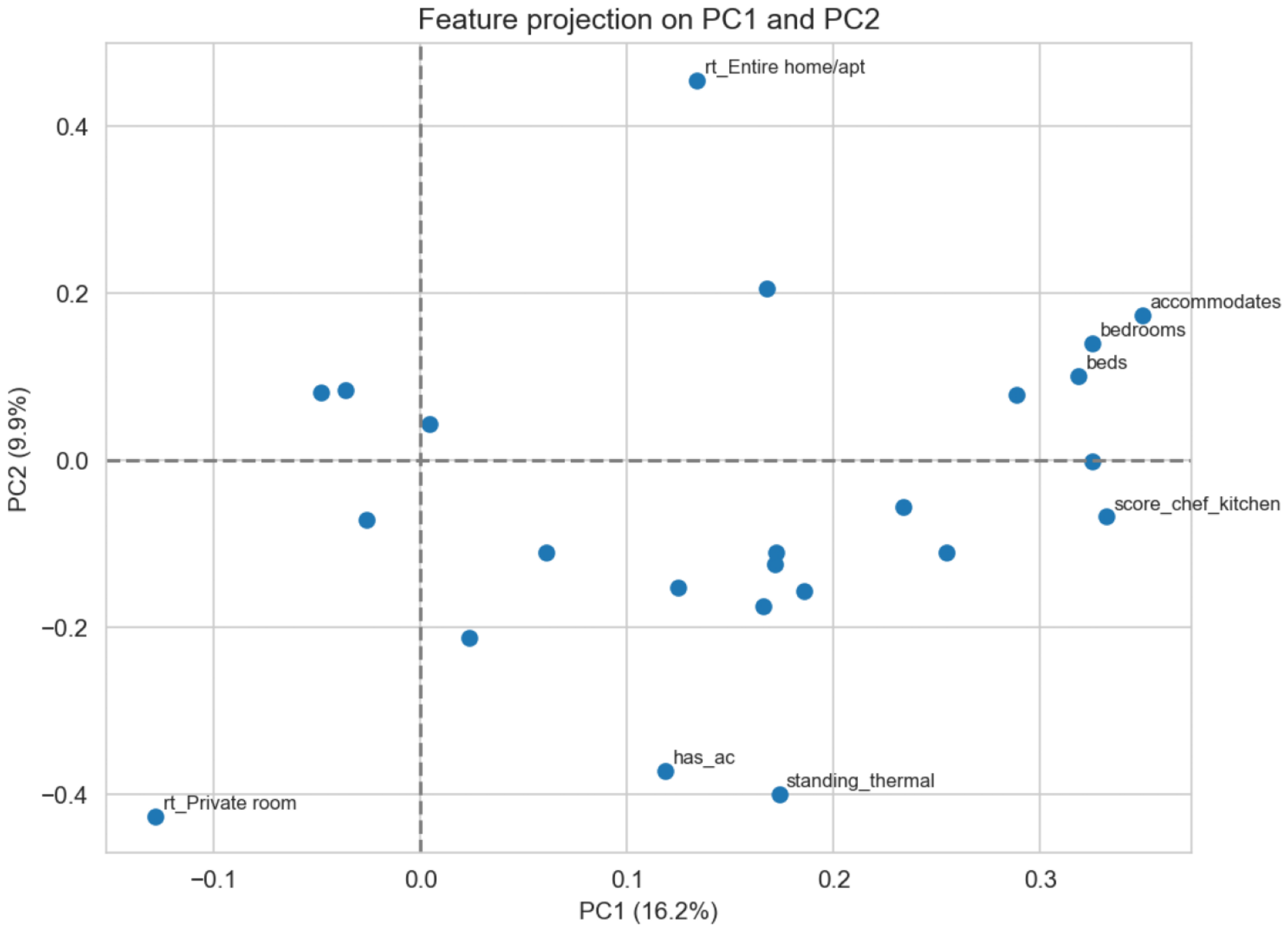

LDA - Test
Classs 0
Classs 1
Classs 2
Classs 3
LD2 (7.6%)
LD1 (91.7%)
40
30
20
10
0
0
5
10
15
20

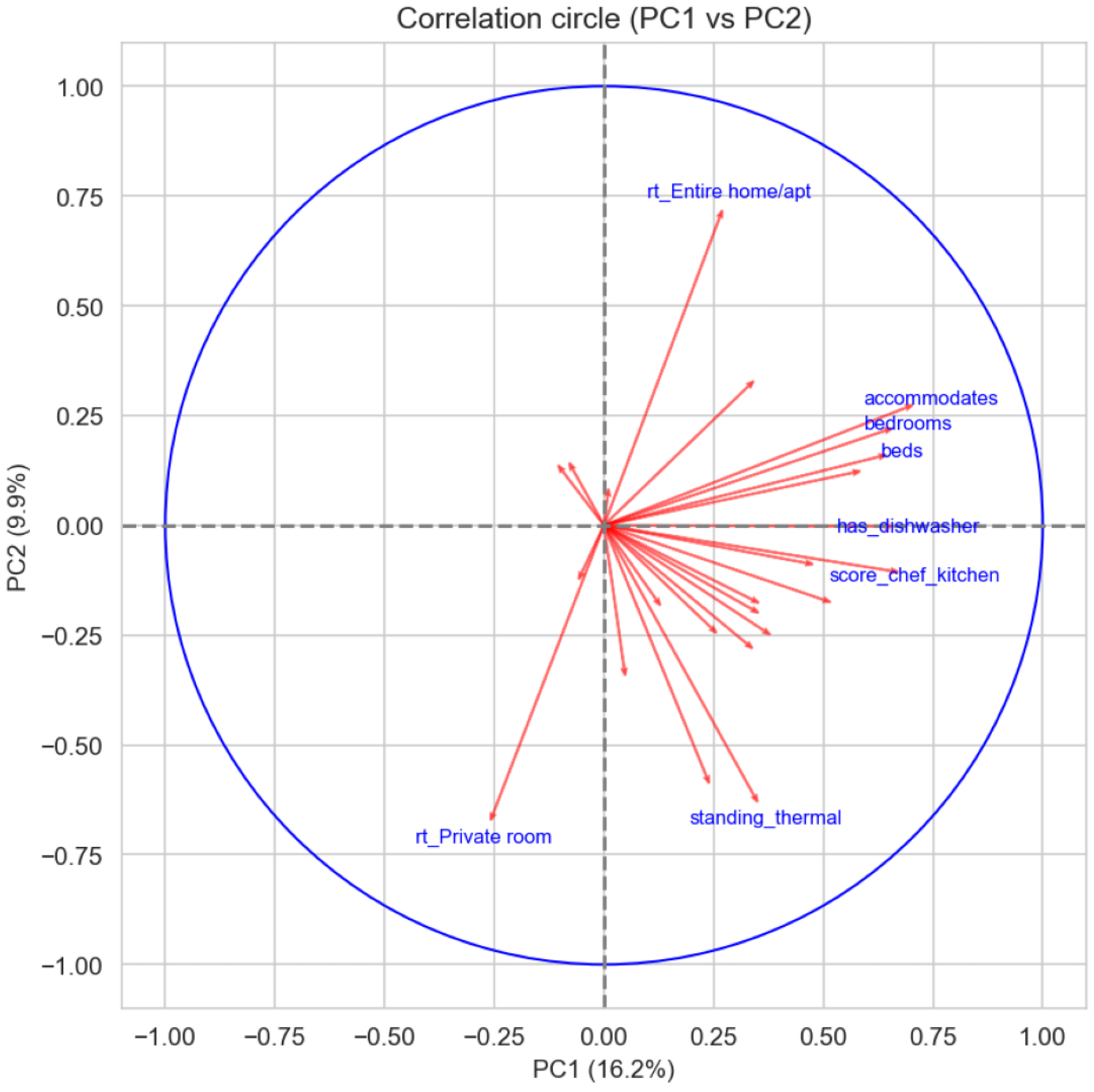

Top 10 standardised weights on LD1
dist_to_center
minimum_nights_avg_ntm
score_secure
score_chef_kitchen
has_ac
has_dishwasher
rt_Entire home/apt
bedrooms
rt_Private room
accommodates
Variables
−0.4
−0.2
0.0
0.2
0.4
0.6
Discriminant weight (standardised features)

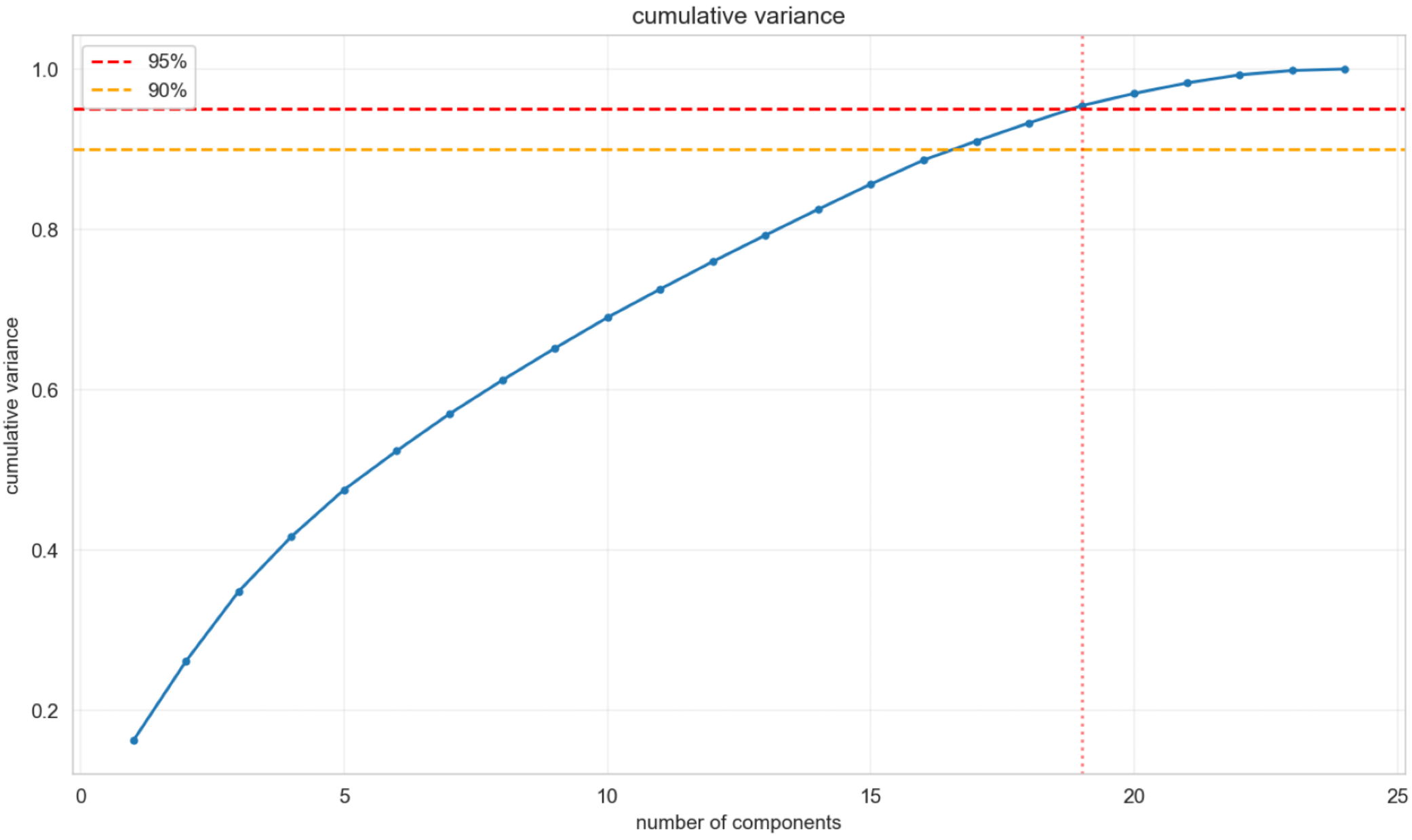

Top 10 standardised weights on LD2
has_washing_machine
accommodates
score_secure
rt_Shared room
minimum_nights_avg_ntm
dist_to_center
rt_Entire home/apt
has_dishwasher
rt_Private room
bathrooms_qty
Variables
−0.4
−0.2
0.0
0.2
0.4
0.6
Discriminant weight (standardised features)


Multi-class ROC Curves (LDA One-vs-Rest)
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Class 0 vs Rest (AUC = 0.82)
Class 1 vs Rest (AUC = 0.67)
Class 2 vs Rest (AUC = 0.67)
Class 3 vs Rest (AUC = 0.83)

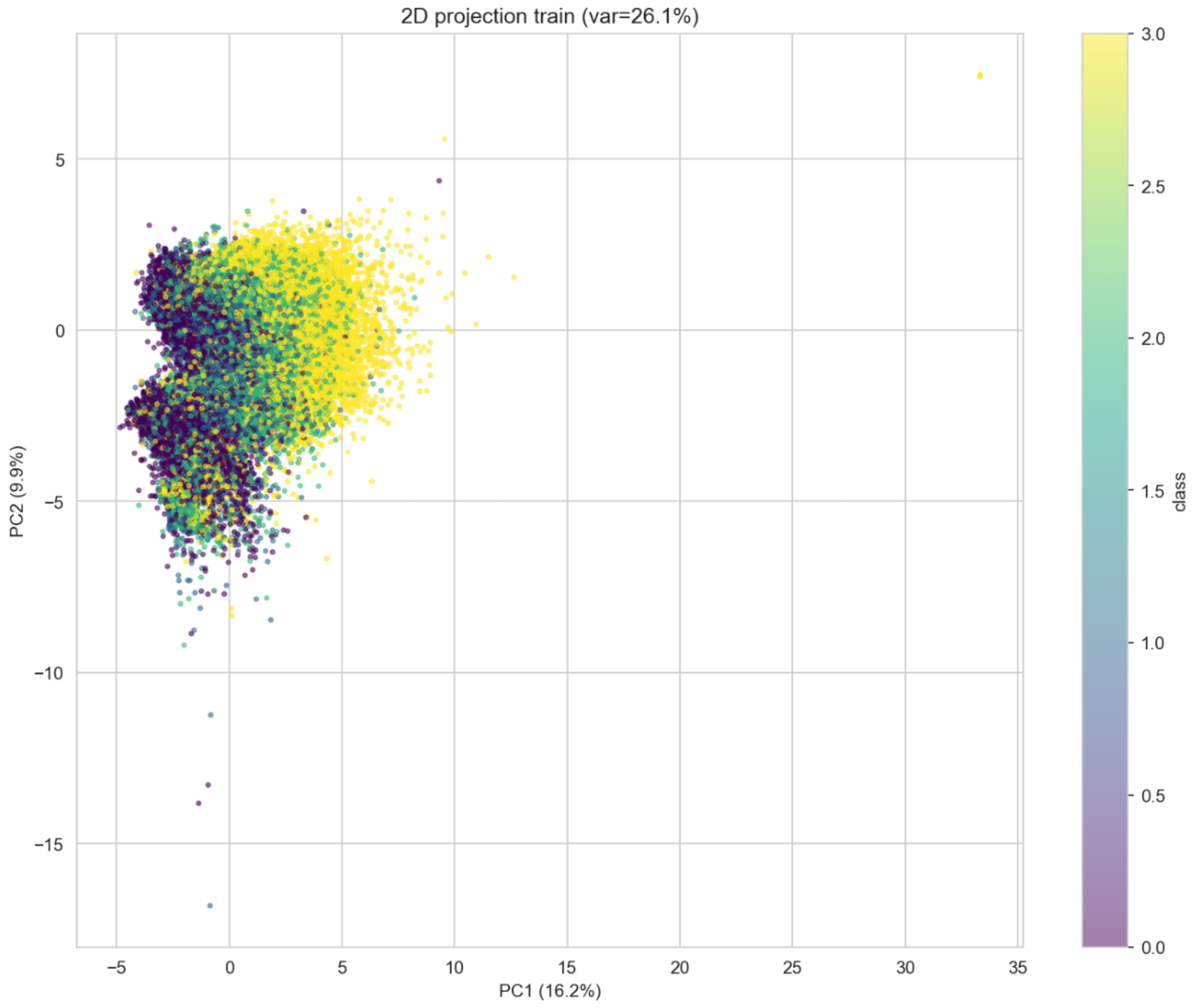

scree plot
explained variance
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
component

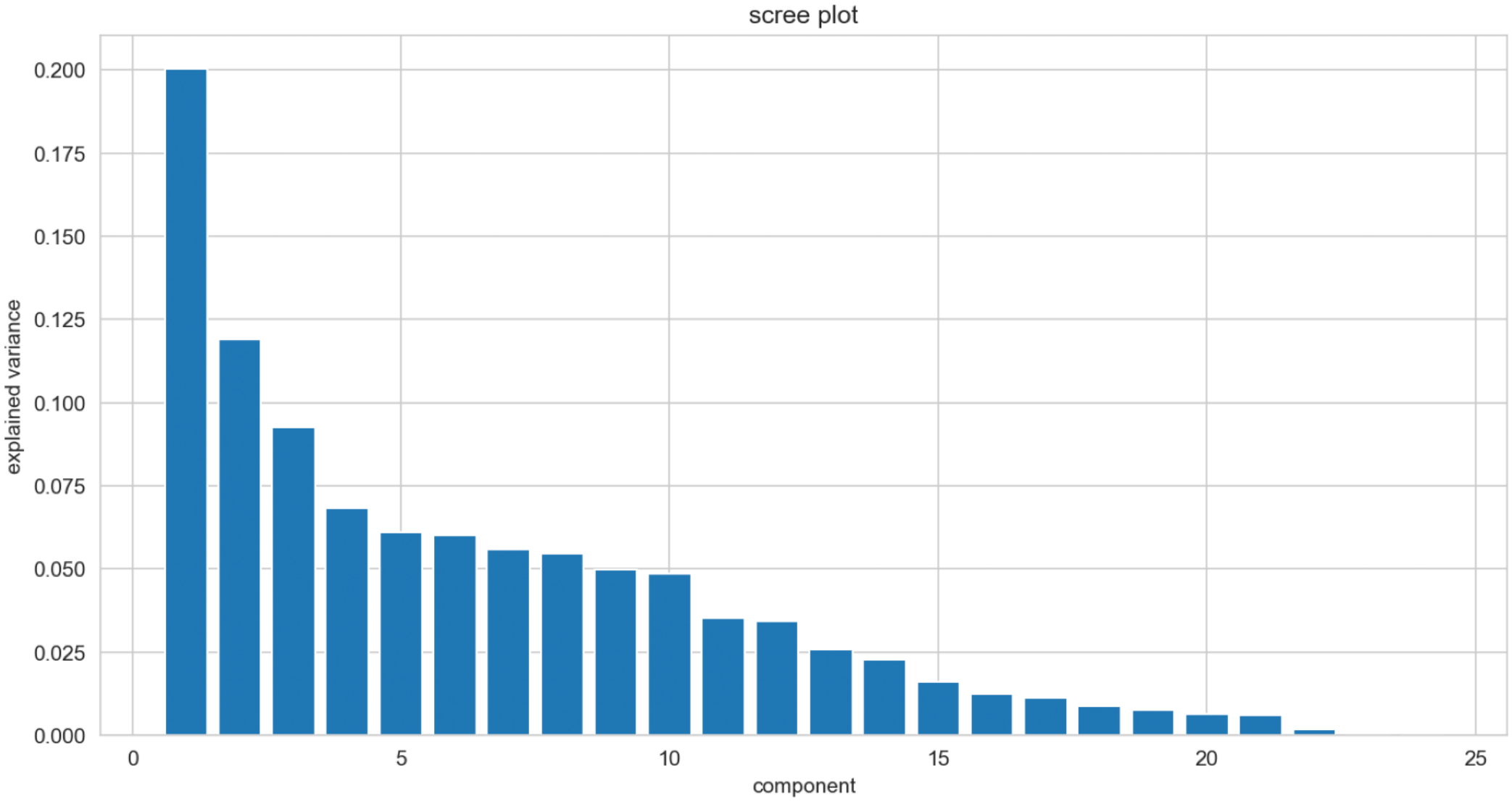

Feature projection on PC1 and PC2
rt_Entire home/apt
accommodates
bedrooms
beds
score_chef_kitchen
has_ac
standing_thermal
rt_Private room
0.4
0.2
0.0
−0.2
−0.4
PC2 (9.9%)
−0.1
0.0
0.1
0.2
0.3
PC1 (16.2%)

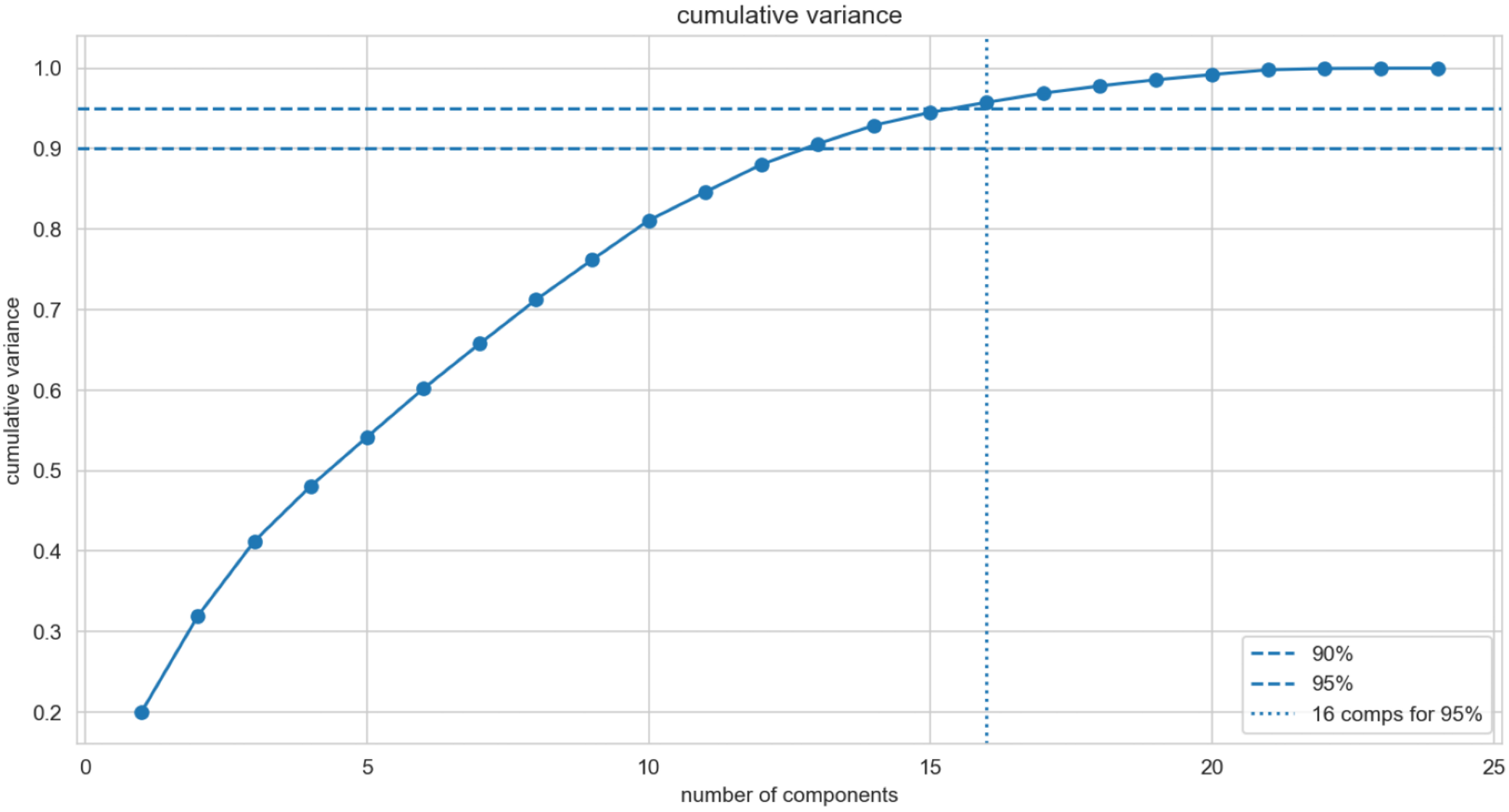

Correlation circle (PC1 vs PC2)
rt_Entire home/apt
accommodates
bedrooms
beds
has_dishwasher
score_chef_kitchen
standing_thermal
rt_Private room
PC1 (16.2%)
PC2 (9.9%)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

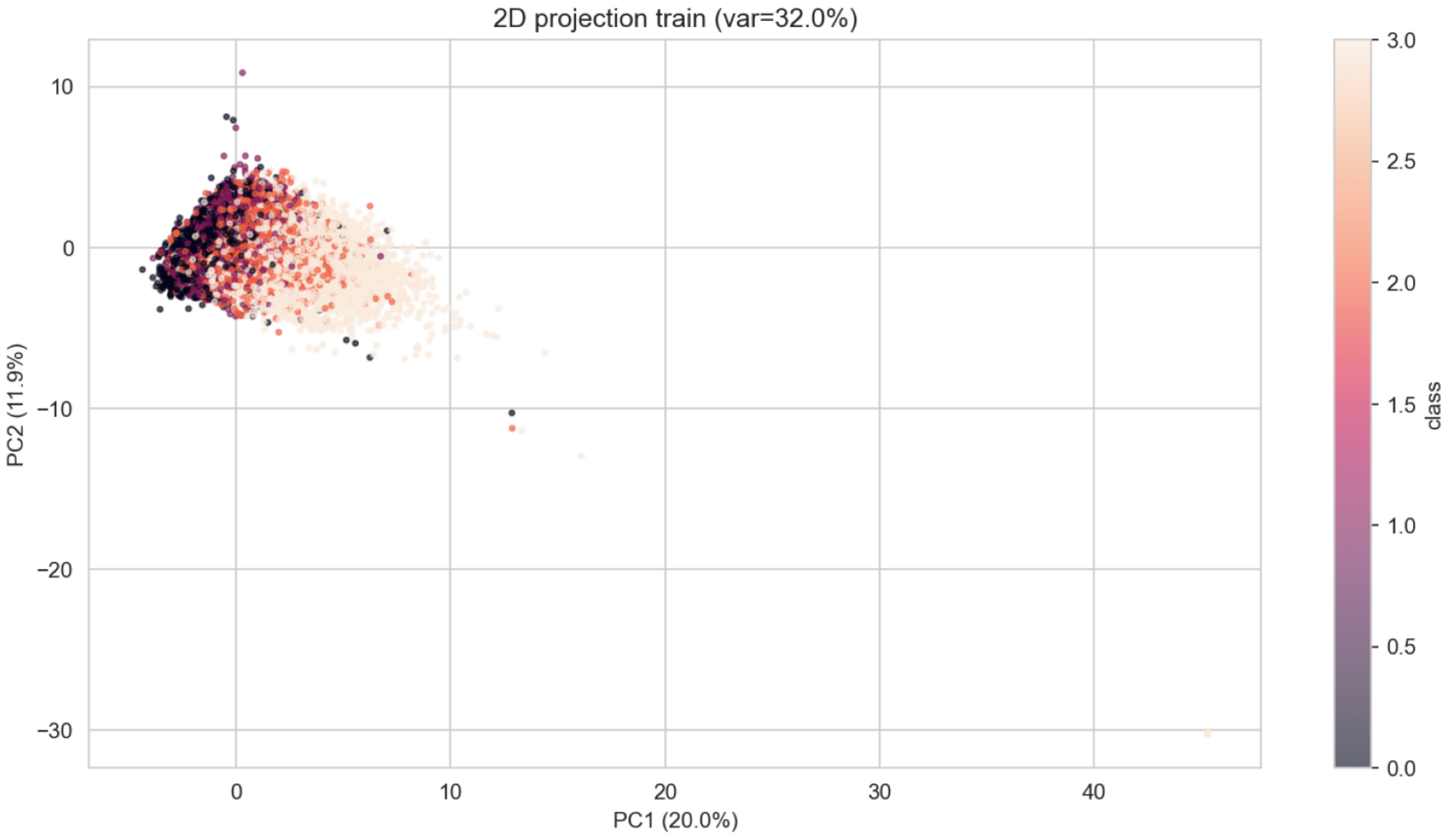

cumulative variance
95%
90%
1.0
0.8
0.6
0.4
0.2
cumulative variance
0
5
10
15
20
25
number of components

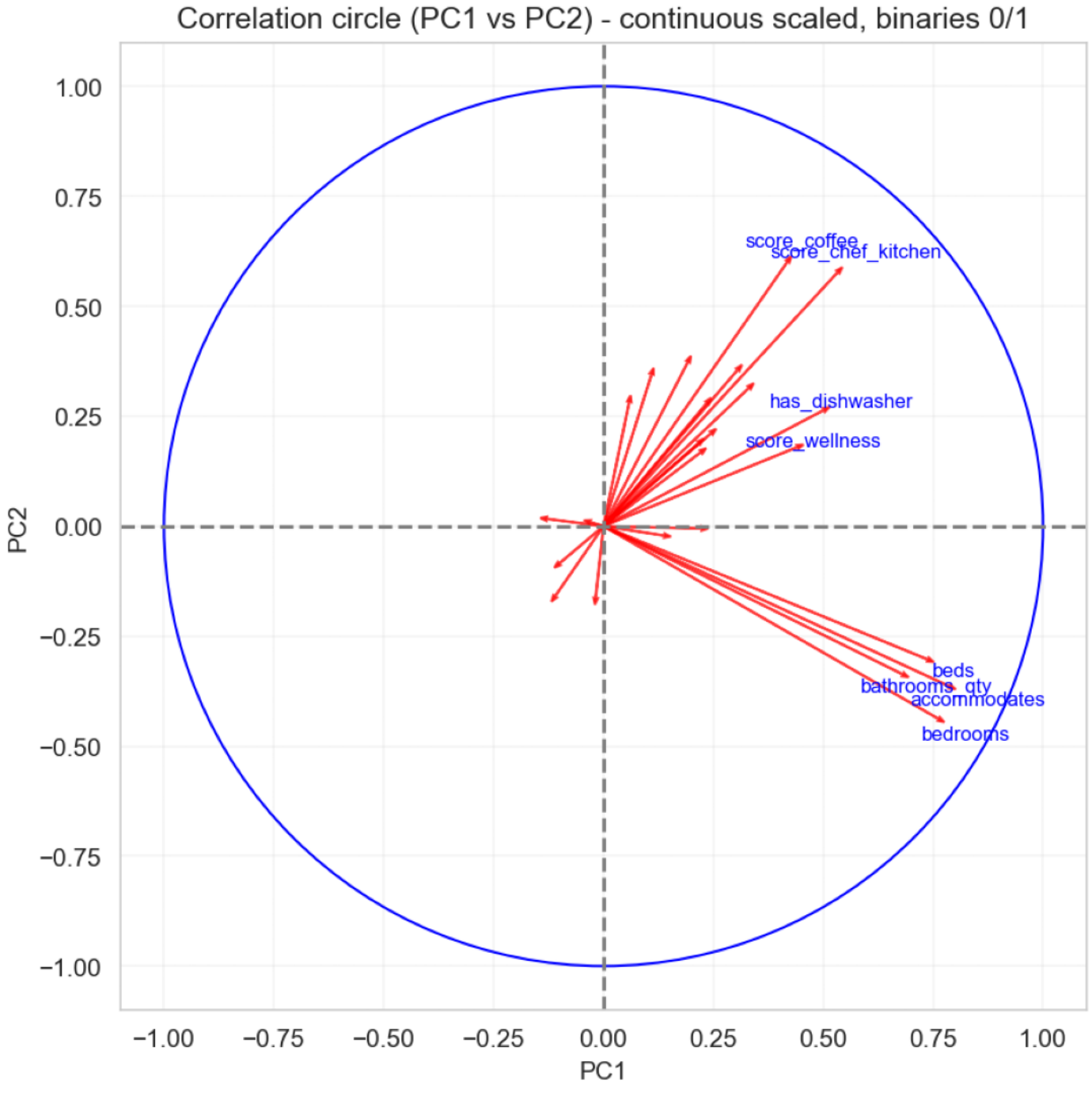
1_ML_Paris / pca_A_05_projection_by_class

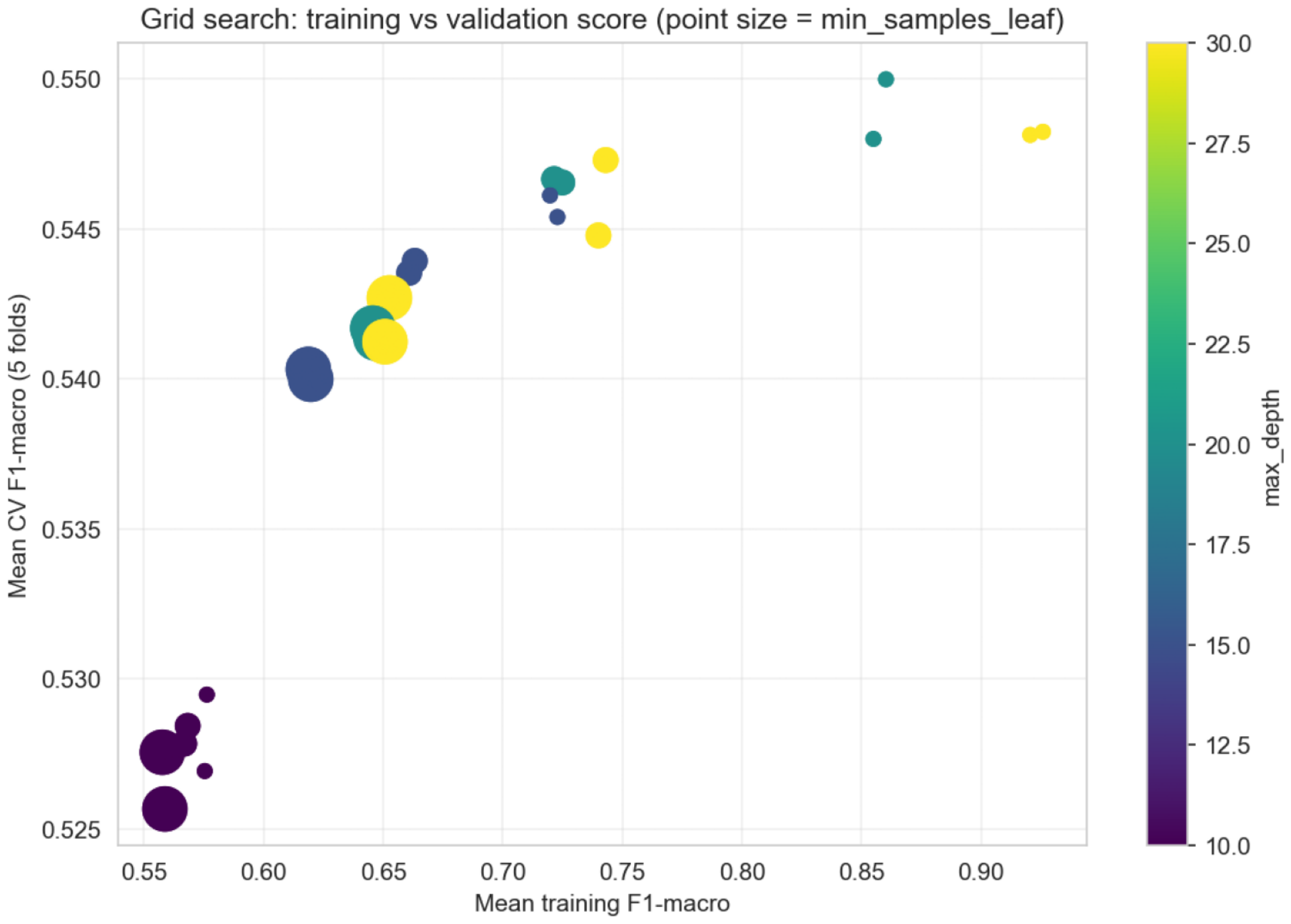


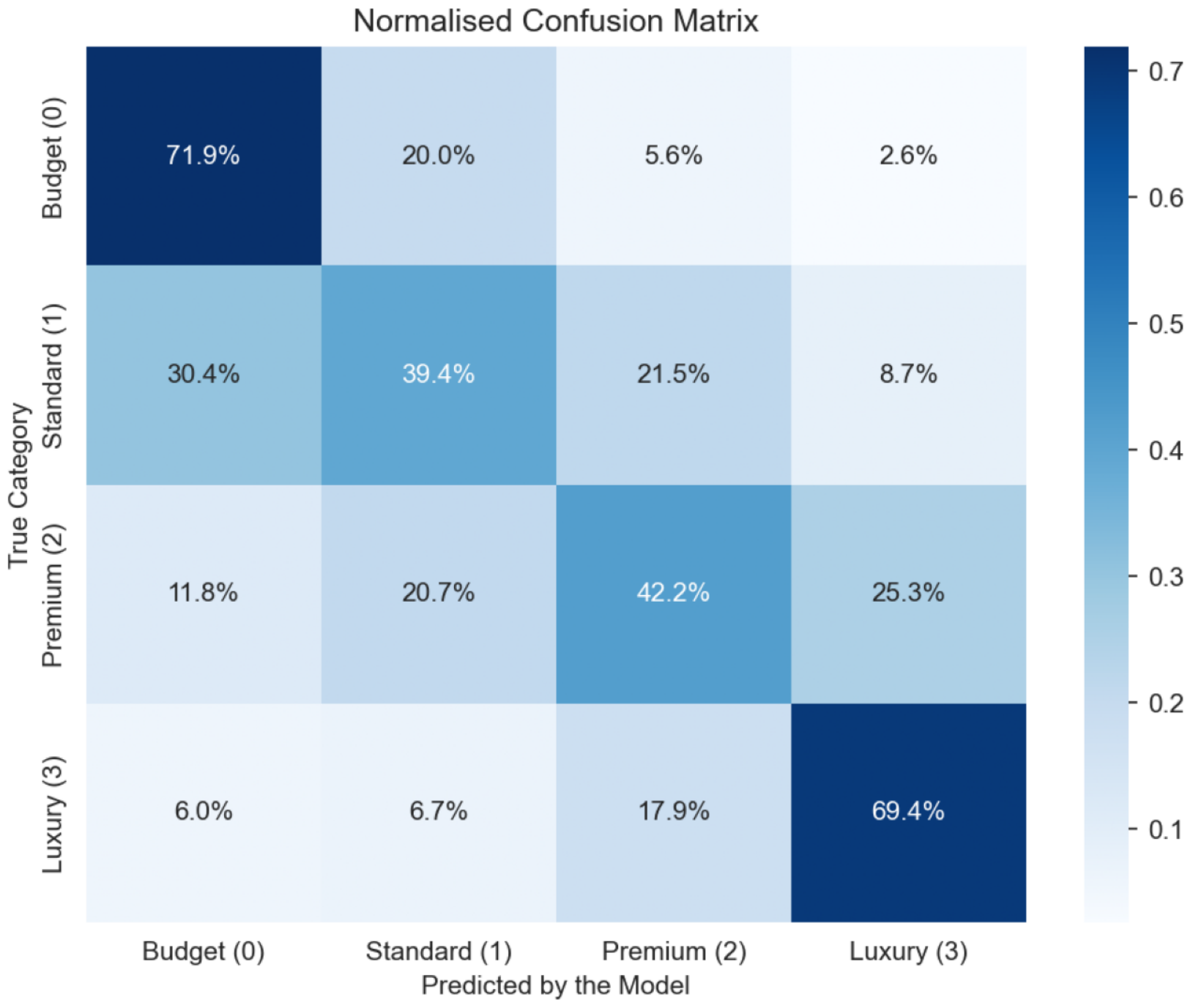

scree plot
explained variance
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
5
10
15
20
25
component

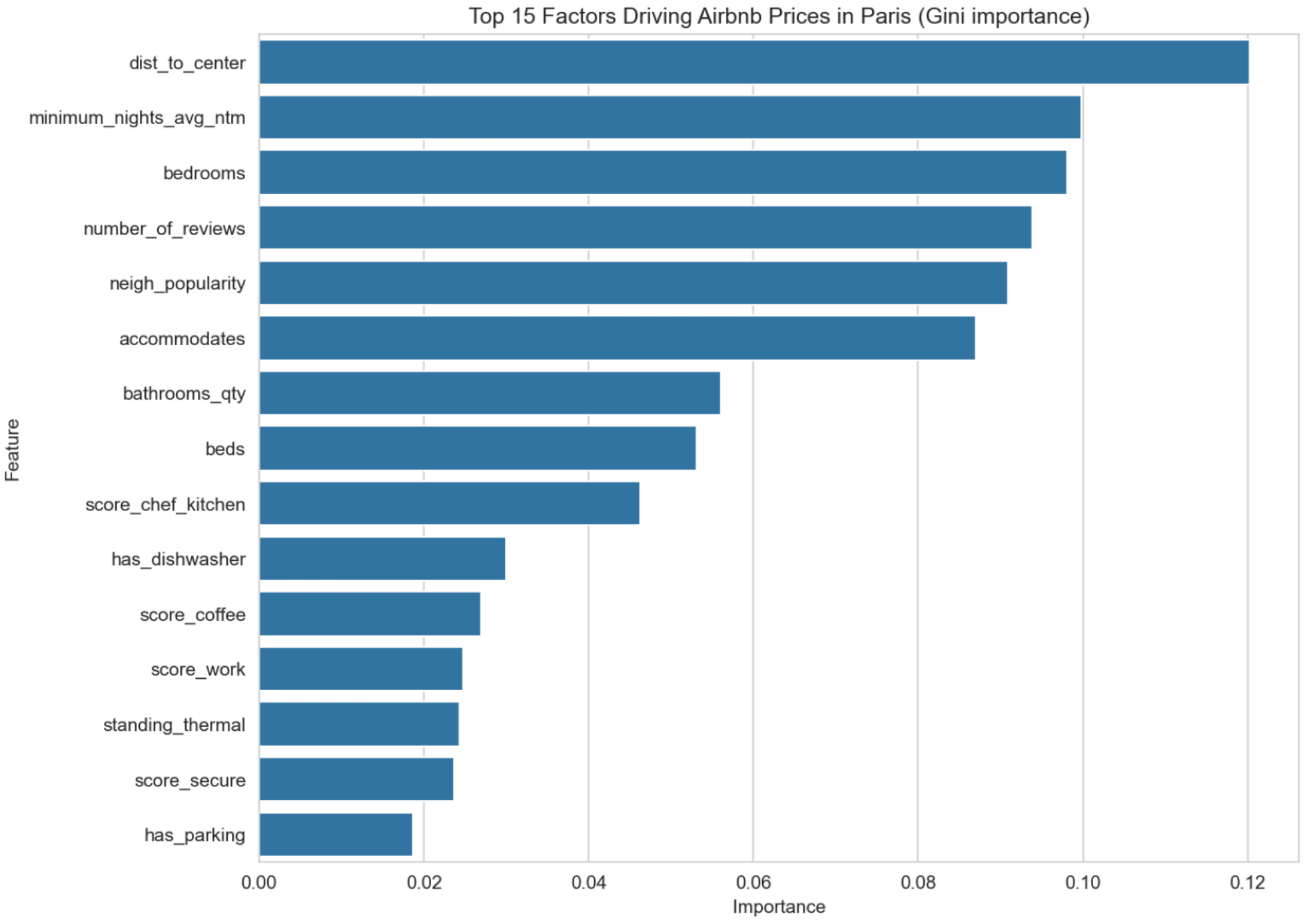

cumulative variance
cumulative variance
number of components
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
5
10
15
20
25
90%
95%
16 comps for 95%

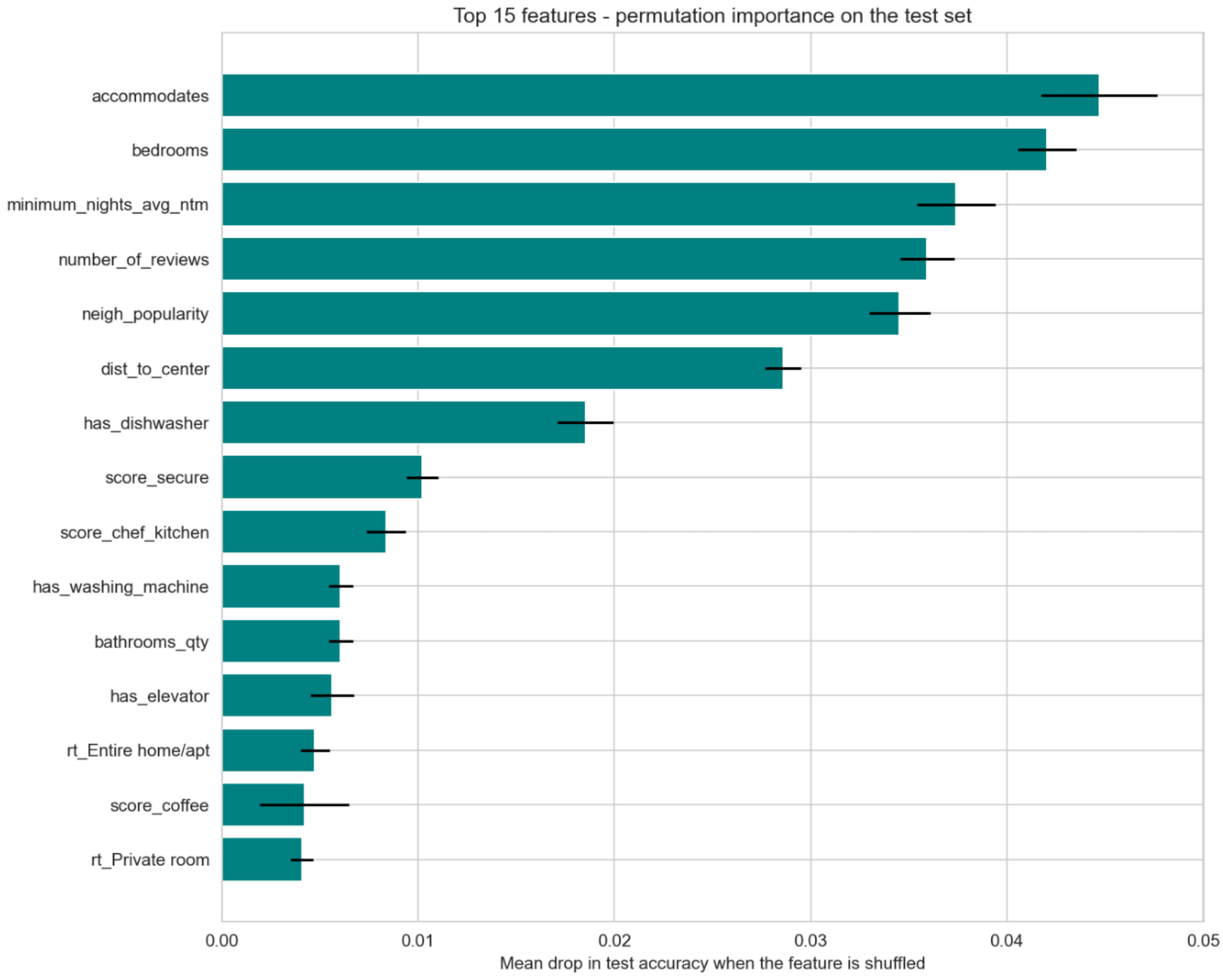

2D projection train (var=32.0%)
PC2 (11.9%)
PC1 (20.0%)
class
10
0
−10
−20
−30
0
10
20
30
40
3.0
2.5
2.0
1.5
1.0
0.5
0.0

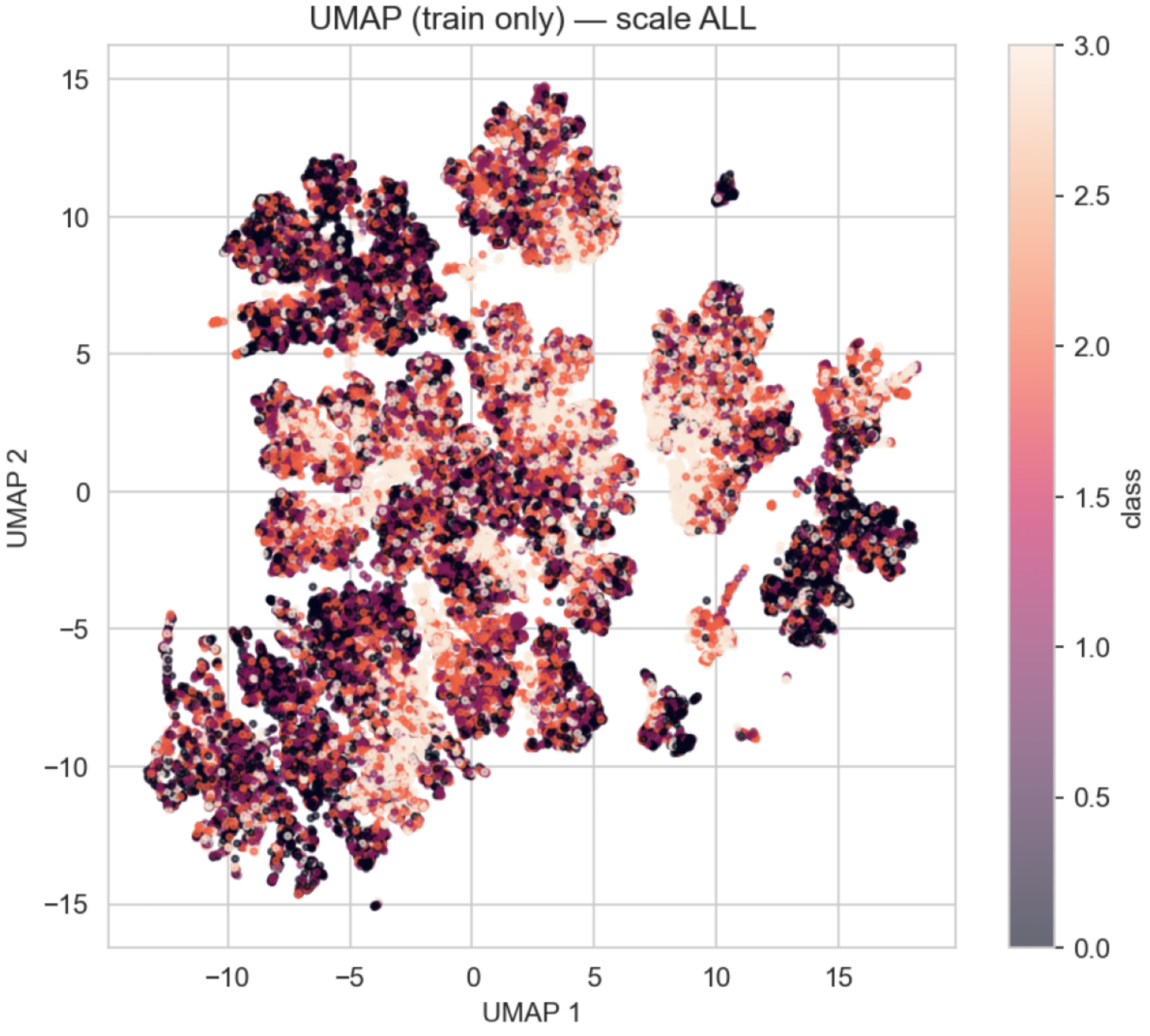

Correlation circle (PC1 vs PC2) - continuous scaled, binaries 0/1
score_coffee
score_chef_kitchen
has_dishwasher
score_wellness
beds
bathrooms
qty
accommodates
bedrooms
PC1
PC2
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

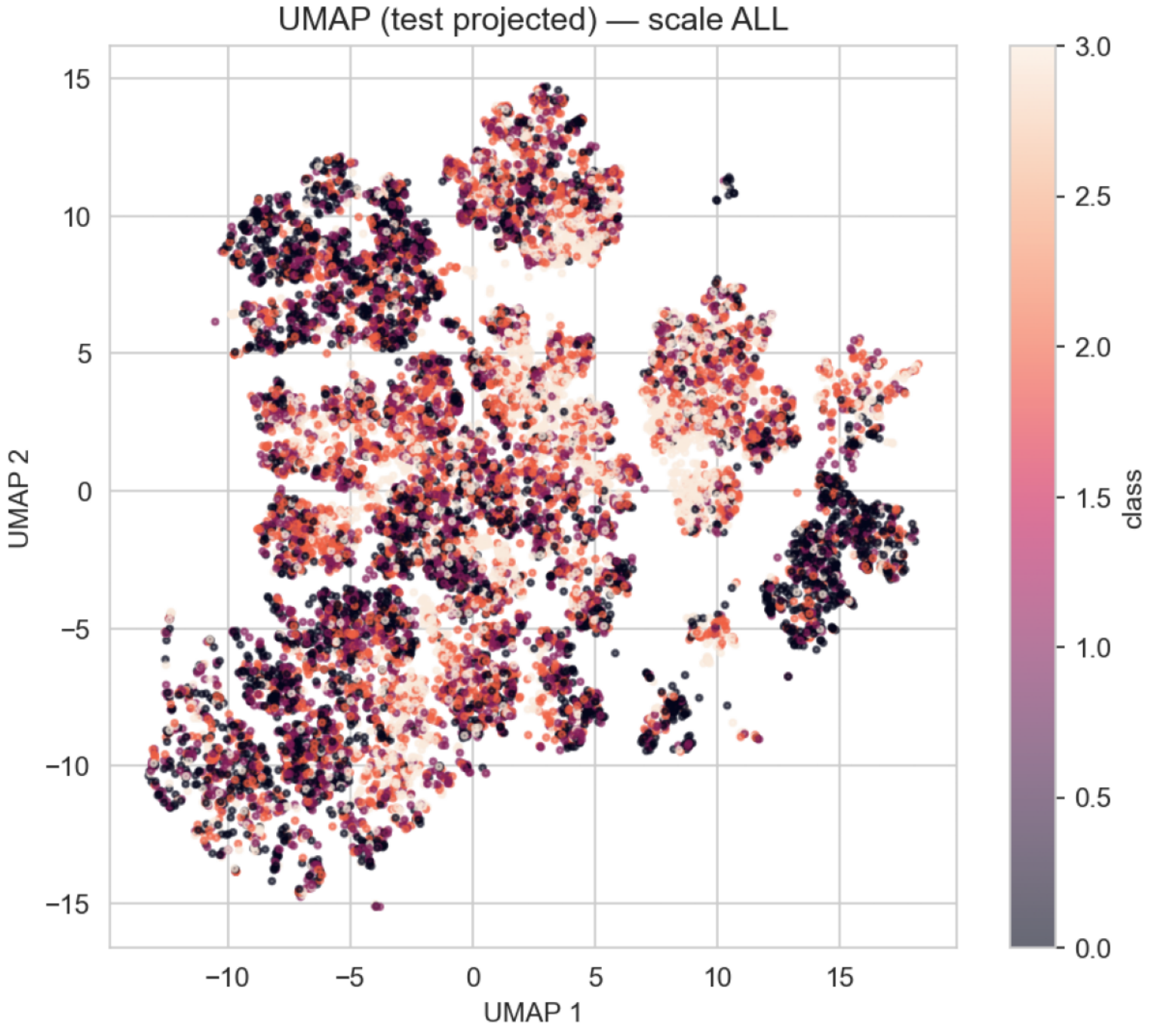

Grid search: training vs validation score (point size = min_samples_leaf)
Mean CV F1-macro (5 folds)
0.550
0.545
0.540
0.535
0.530
0.525
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
Mean training F1-macro
max_depth
30.0
27.5
25.0
22.5
20.0
17.5
15.0
12.5
10.0

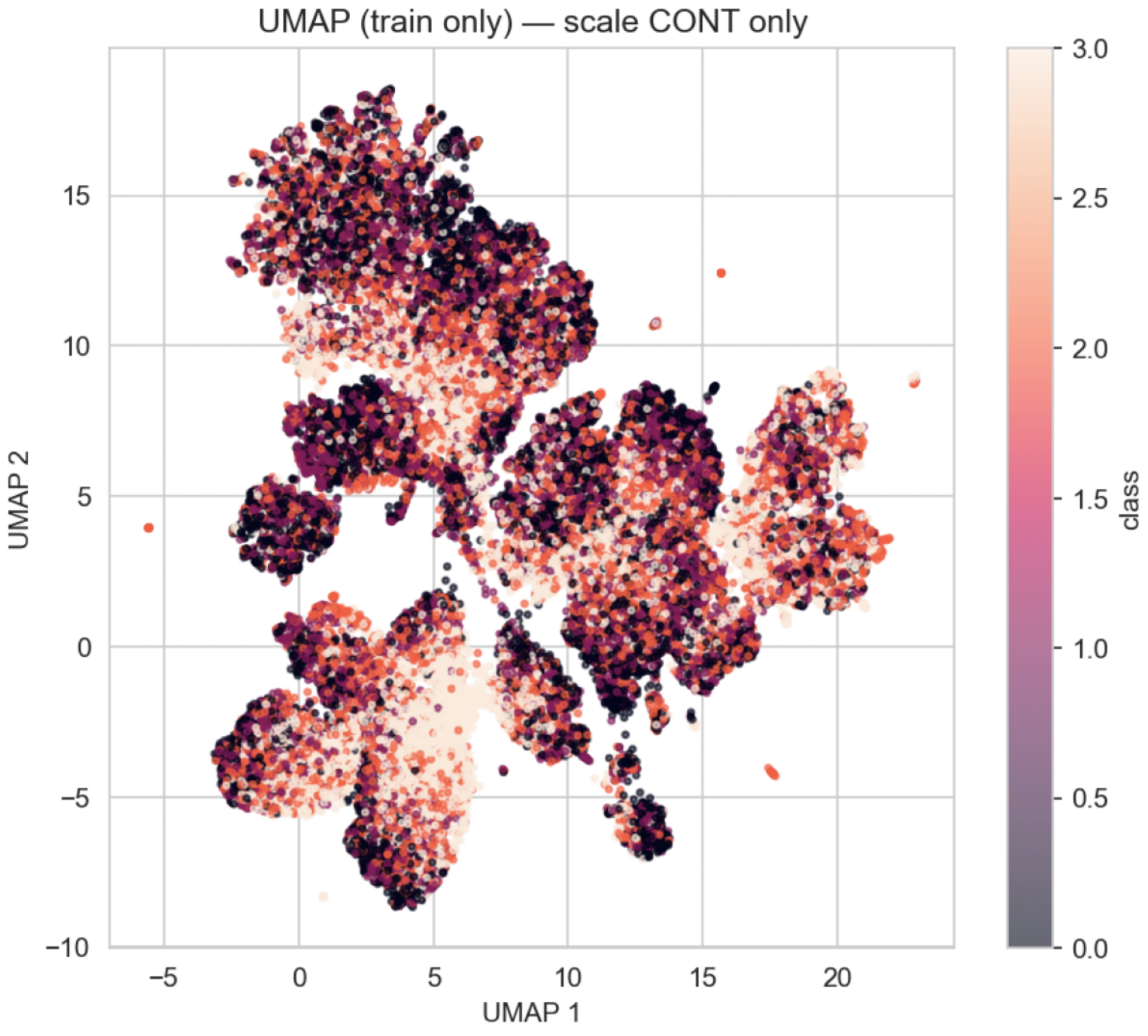

Normalised Confusion Matrix
True Category
Budget (0)
Standard (1)
Premium (2)
Luxury (3)
71.9%
20.0%
5.6%
2.6%
30.4%
39.4%
21.5%
8.7%
11.8%
20.7%
42.2%
25.3%
6.0%
6.7%
17.9%
69.4%
Budget (0)
Standard (1)
Premium (2)
Luxury (3)
Predicted by the Model
0.7
0.6
0.5
0.4
0.3
0.2
0.1

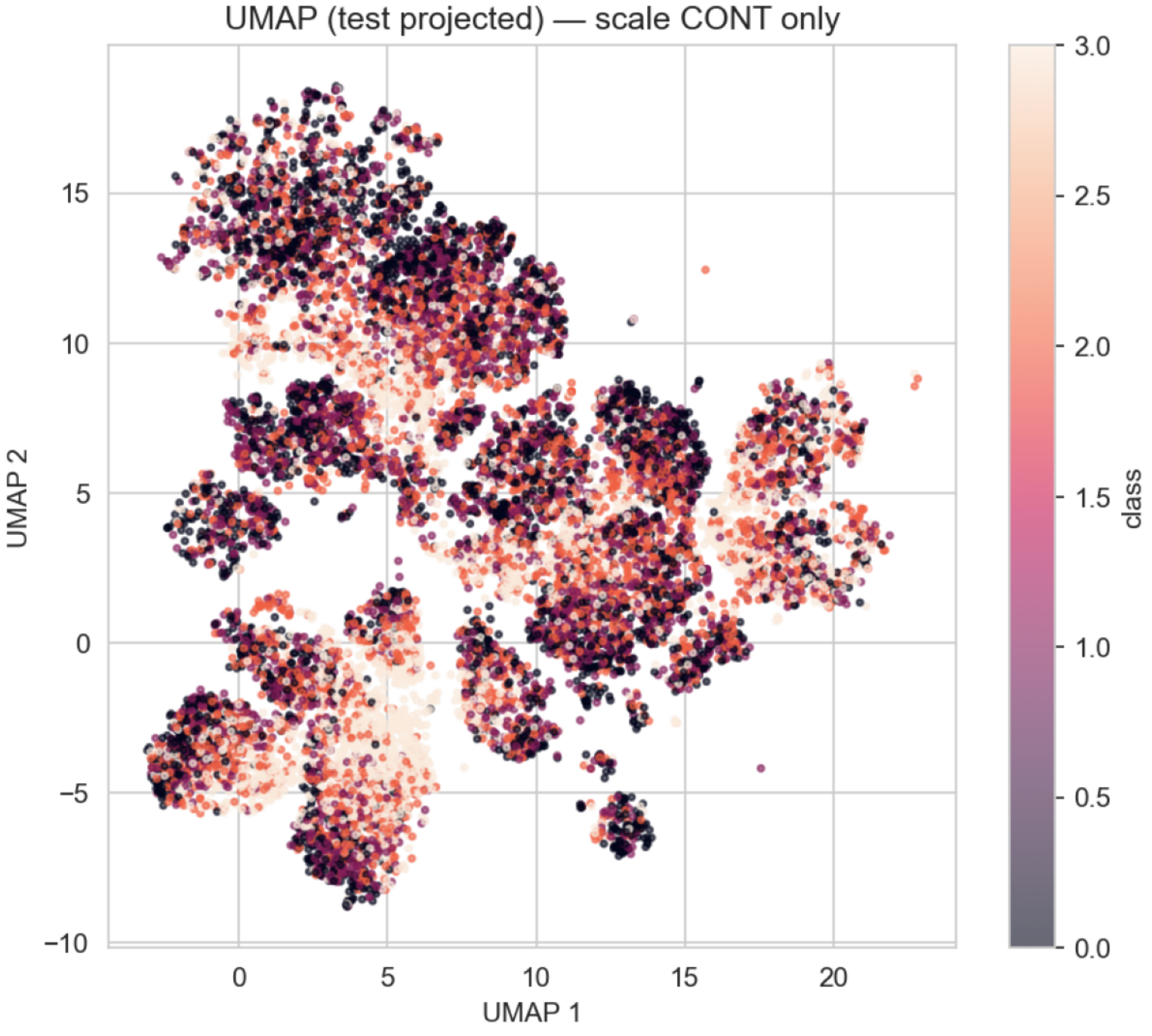

Top 15 Factors Driving Airbnb Prices in Paris (Gini importance)
dist_to_center
minimum_nights_avg_ntm
bedrooms
number_of_reviews
neigh_popularity
accommodates
bathrooms_qty
beds
score_chef_kitchen
has_dishwasher
score_coffee
score_work
standing_thermal
score_secure
has_parking
Feature
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Importance

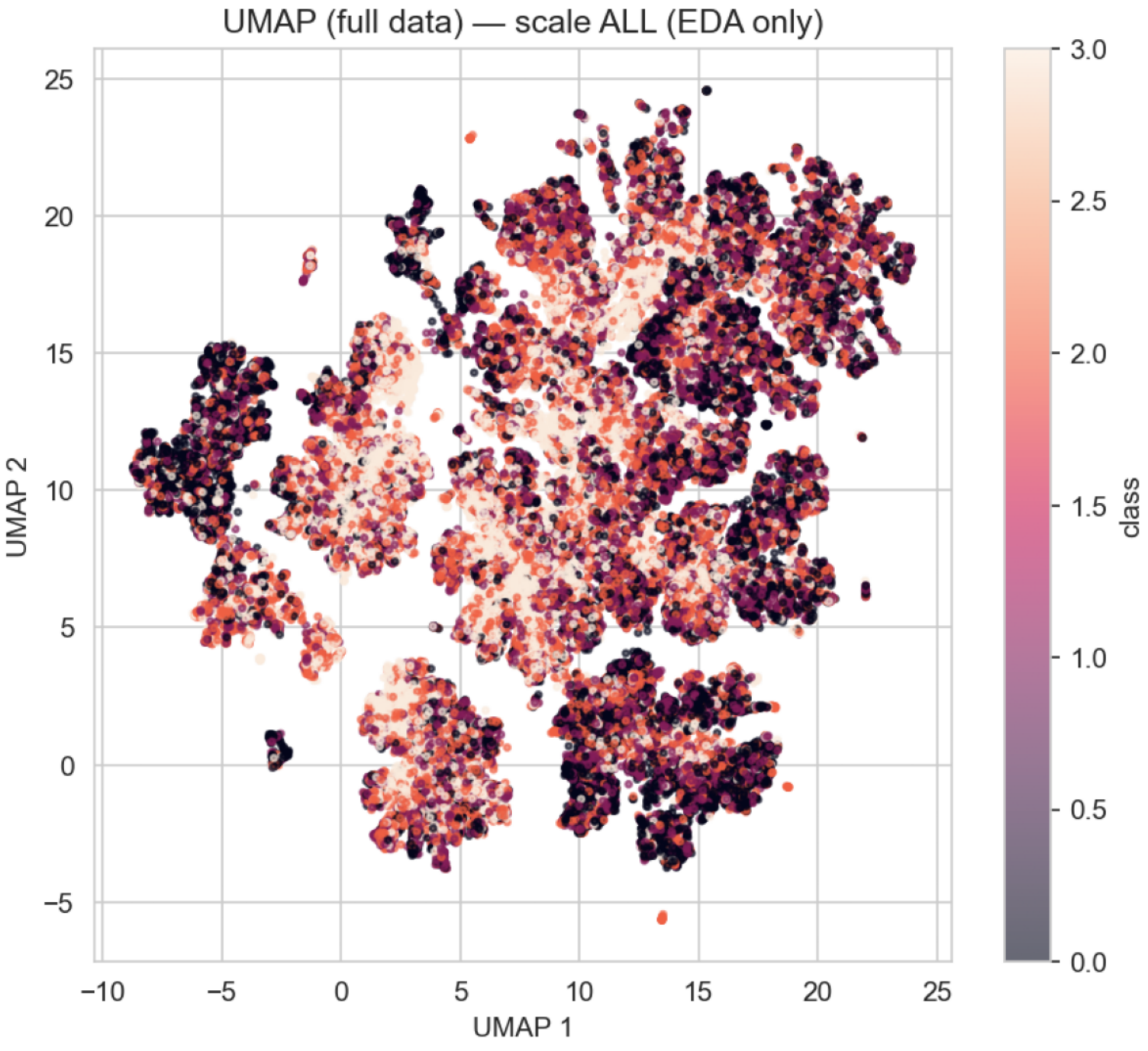

Top 15 features - permutation importance on the test set
accommodates
bedrooms
minimum_nights_avg_ntm
number_of_reviews
neigh_popularity
dist_to_center
has_dishwasher
score_secure
score_chef_kitchen
has_washing_machine
bathrooms_qty
has_elevator
rt_Entire home/apt
score_coffee
rt_Private room
0.00
0.01
0.02
0.03
0.04
0.05
Mean drop in test accuracy when the feature is shuffled

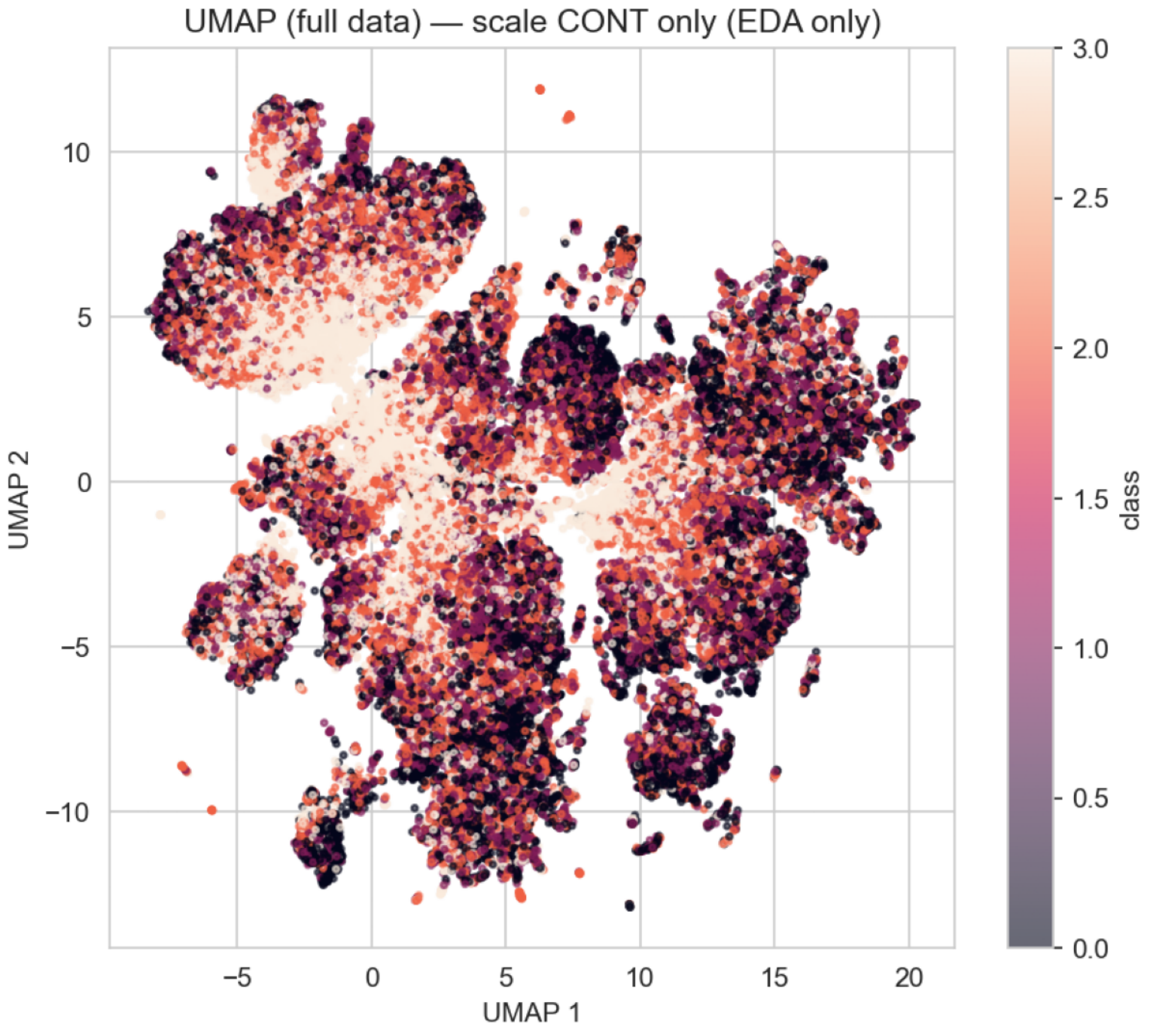

UMAP (train only) — scale ALL
UMAP 2
UMAP 1
class
15
10
5
0
−5
−10
−15
−10
−5
0
5
10
15
3.0
2.5
2.0
1.5
1.0
0.5
0.0

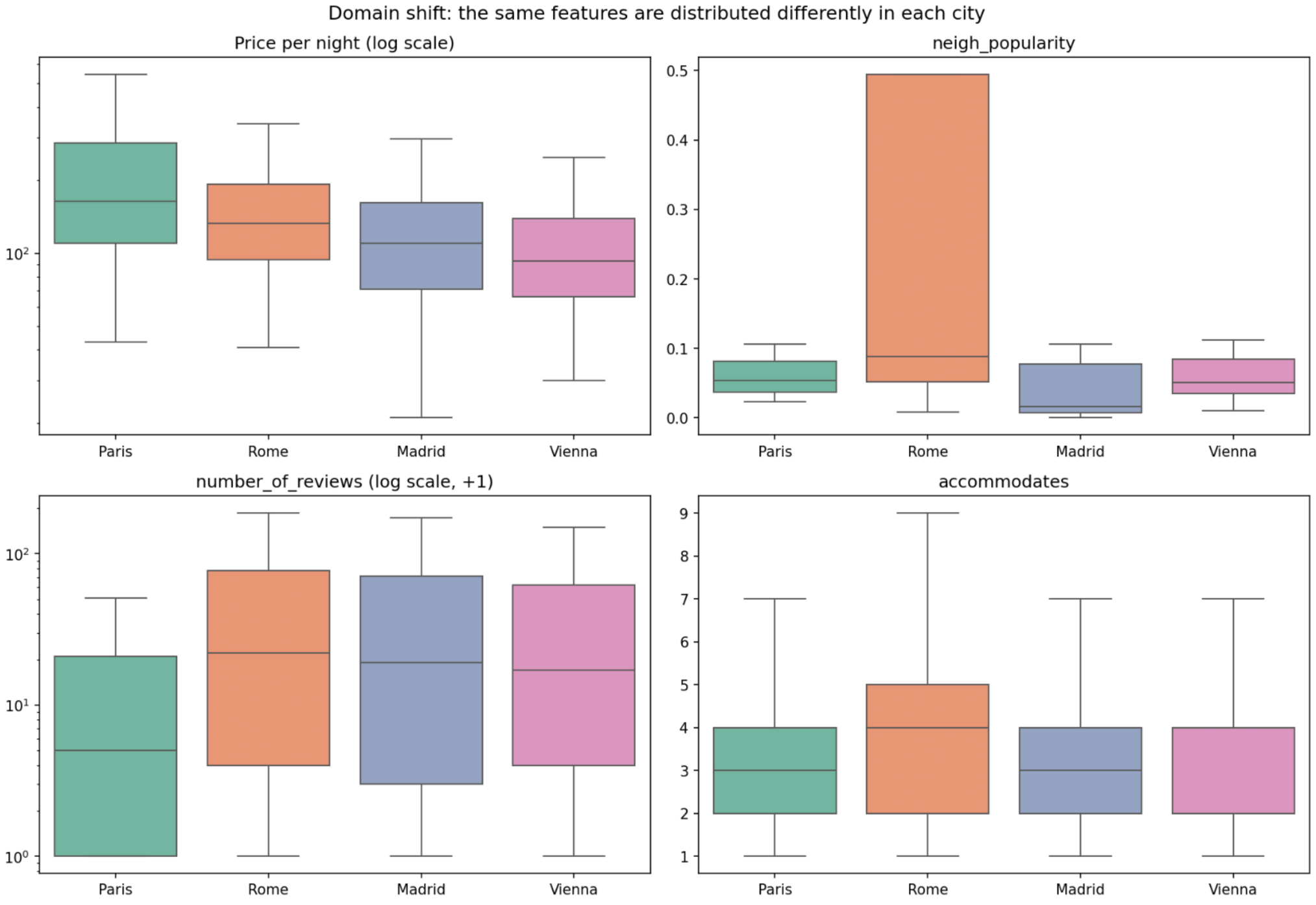

UMAP (test projected) — scale ALL
UMAP 2
UMAP 1
class
15
10
5
0
−5
−10
−15
−10
−5
0
5
10
15
3.0
2.5
2.0
1.5
1.0
0.5
0.0

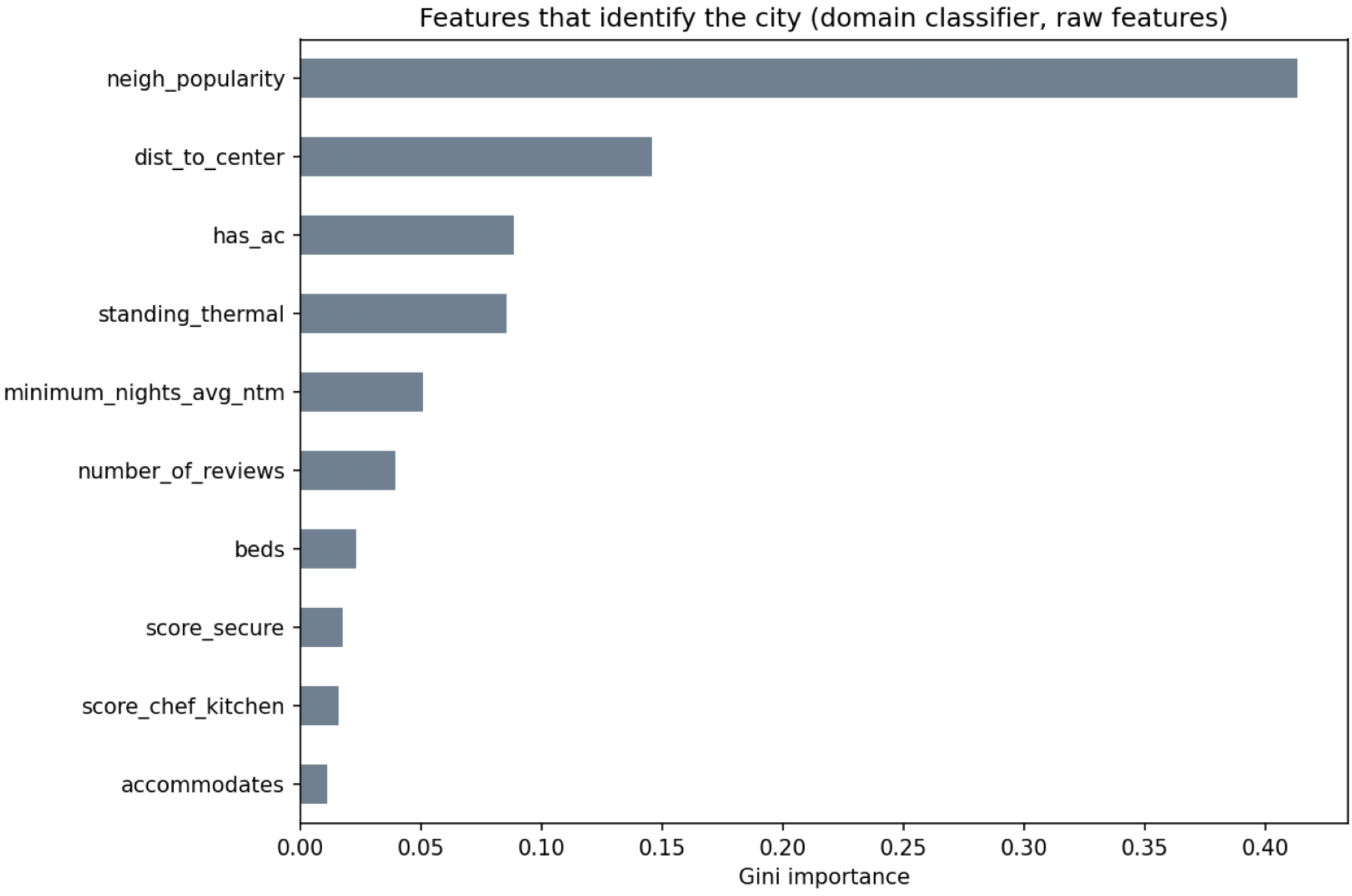

UMAP (train only) — scale CONT only
UMAP 2
UMAP 1
class
15
10
5
0
−5
−10
−5
0
5
10
15
20
3.0
2.5
2.0
1.5
1.0
0.5
0.0

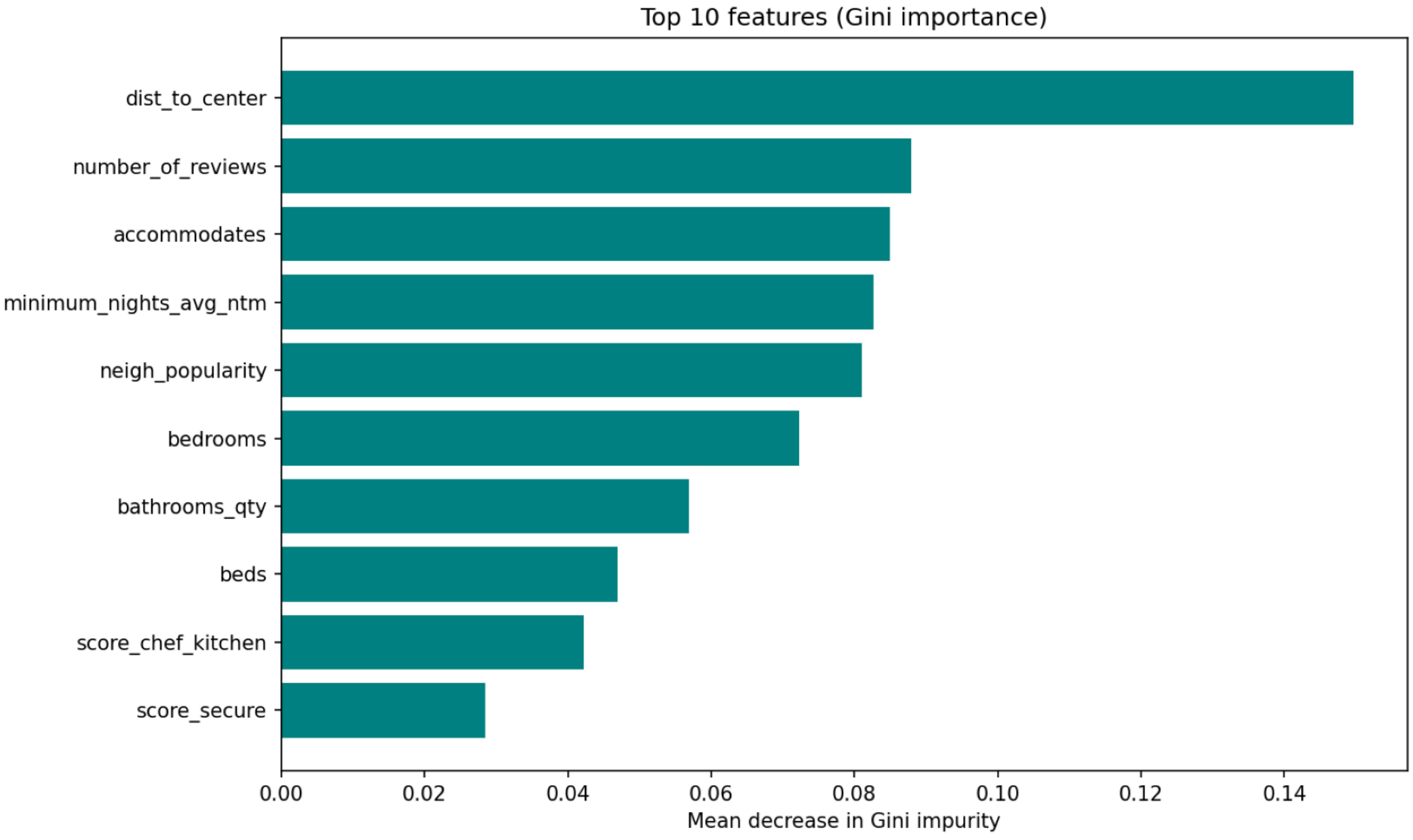

UMAP (test projected) — scale CONT only
UMAP 2
UMAP 1
class
15
10
5
0
−5
−10
0
5
10
15
20
3.0
2.5
2.0
1.5
1.0
0.5
0.0

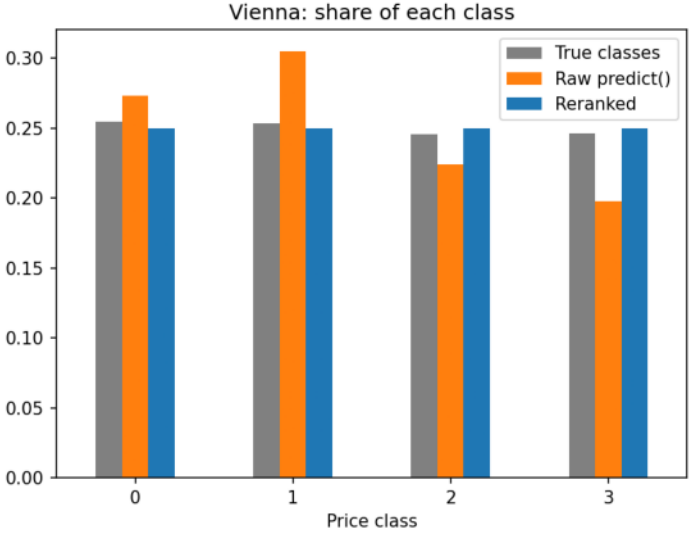

UMAP (full data) — scale ALL (EDA only)
UMAP 2
UMAP 1
class
25
20
15
10
5
0
−5
−10
−5
0
5
10
15
20
25
3.0
2.5
2.0
1.5
1.0
0.5
0.0

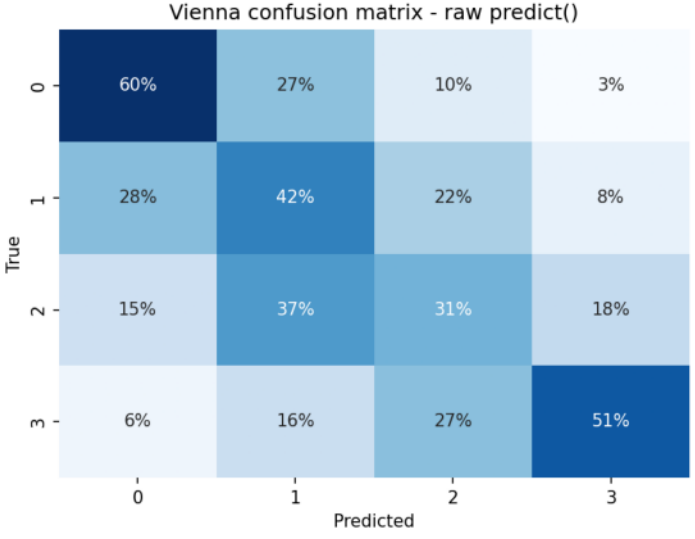

UMAP (full data) — scale CONT only (EDA only)
UMAP 2
UMAP 1
10
5
0
−5
−10
−5
0
5
10
15
20
class
3.0
2.5
2.0
1.5
1.0
0.5
0.0

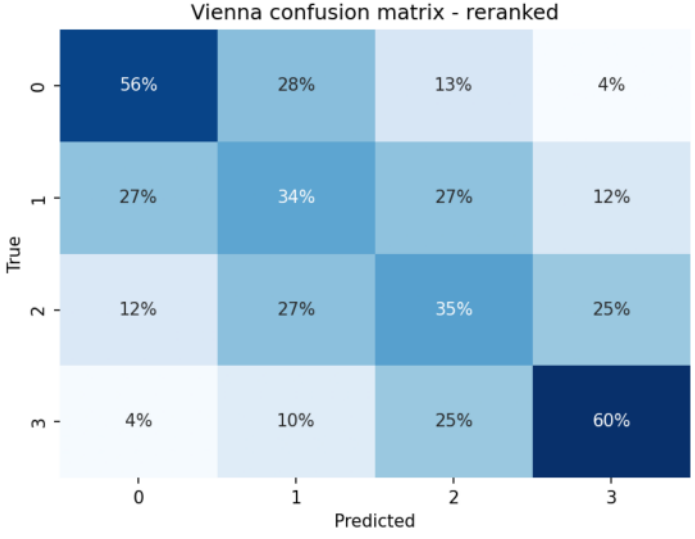

Domain shift: the same features are distributed differently in each city
Price per night (log scale)
neigh_popularity
number_of_reviews (log scale, +1)
accommodates
Paris
Rome
Madrid
Vienna


Features that identify the city (domain classifier, raw features)
neigh_popularity
dist_to_center
has_ac
standing_thermal
minimum_nights_avg_ntm
number_of_reviews
beds
score_secure
score_chef_kitchen
accommodates
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Gini importance

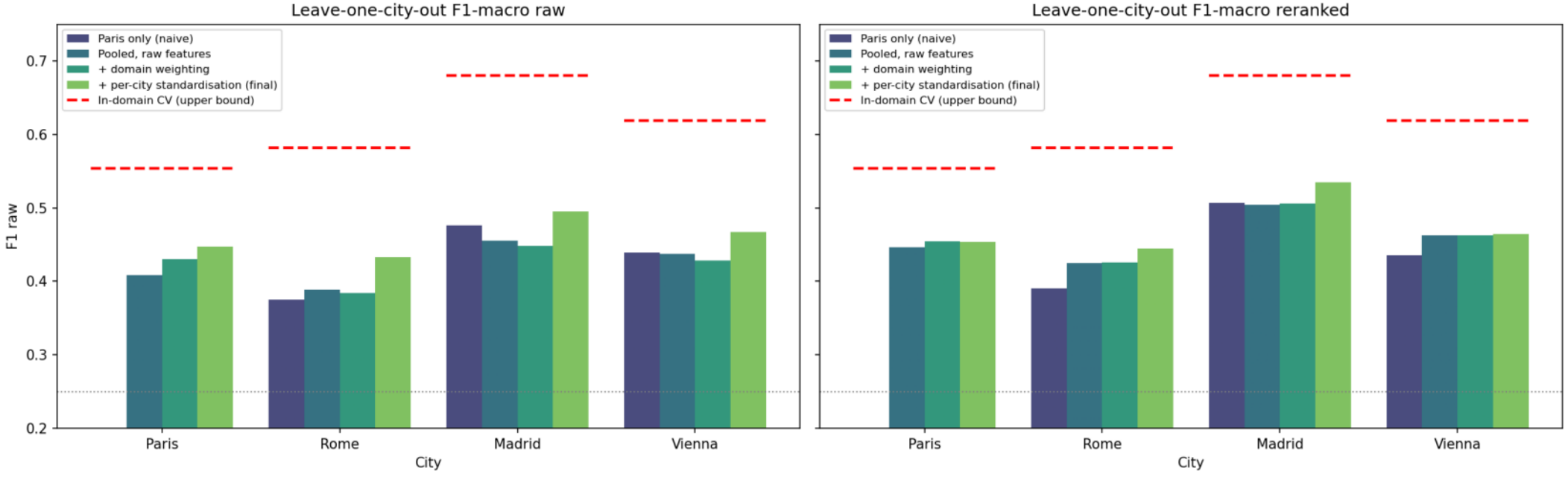

Top 10 features (Gini importance)
dist_to_center
number_of_reviews
accommodates
minimum_nights_avg_ntm
neigh_popularity
bedrooms
bathrooms_qty
beds
score_chef_kitchen
score_secure
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Mean decrease in Gini impurity

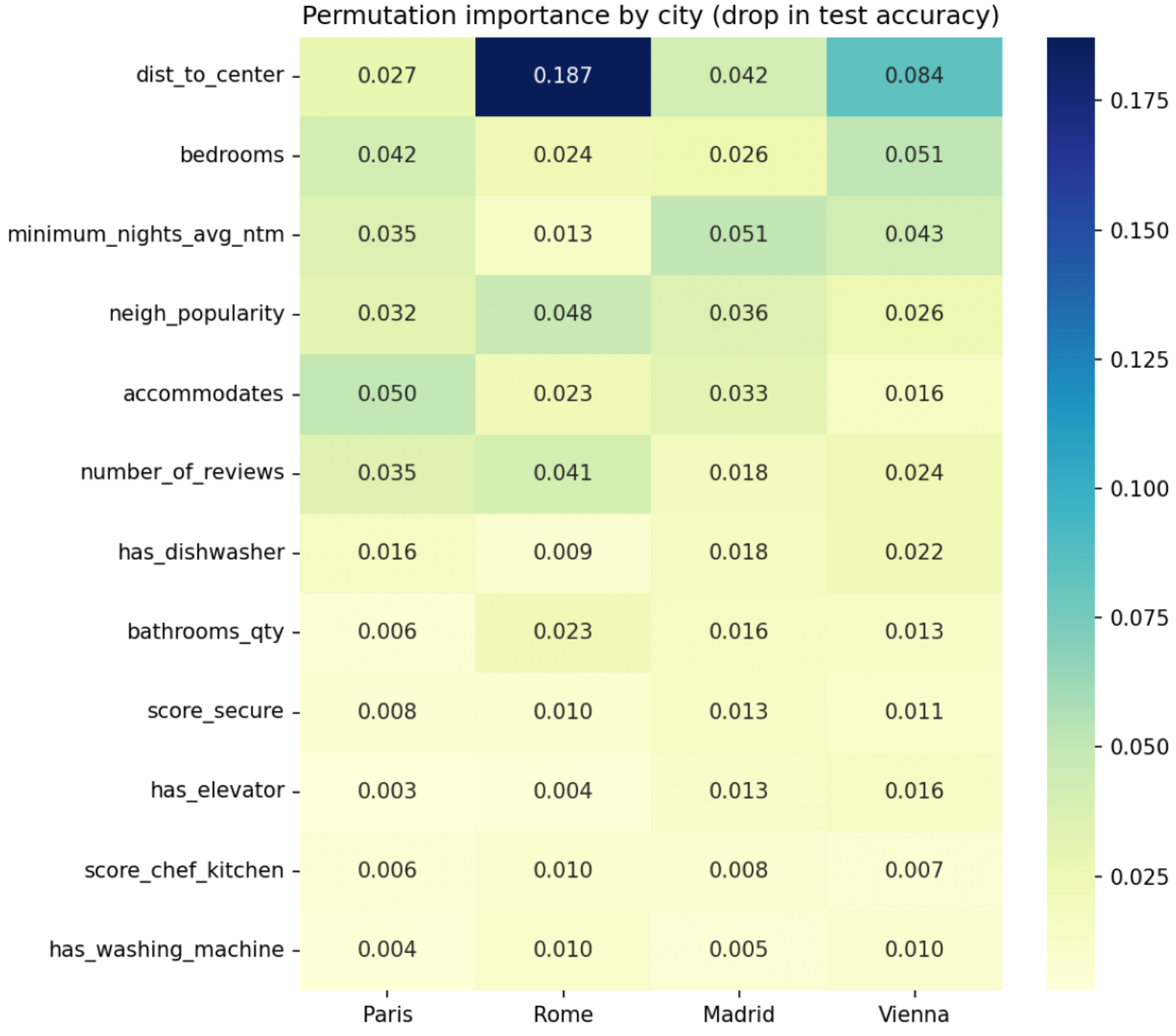

Vienna: share of each class
True classes
Raw predict()
Reranked
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3
Price class

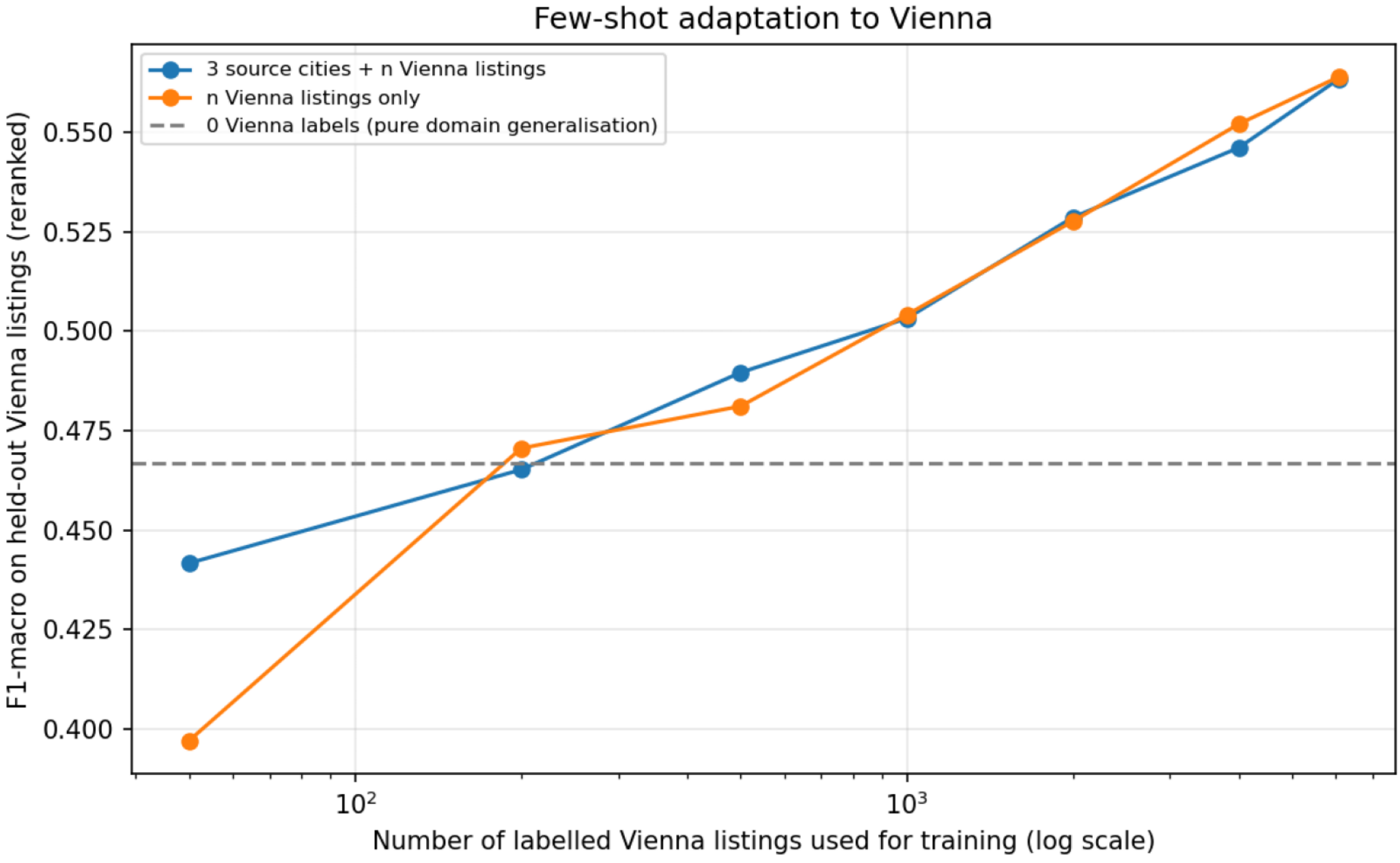

Vienna confusion matrix - raw predict()
True
0
1
2
3
60%
27%
10%
3%
28%
42%
22%
8%
15%
37%
31%
18%
6%
16%
27%
51%
Predicted


Vienna confusion matrix - reranked
True
0
1
2
3
56%
28%
13%
4%
27%
34%
27%
12%
12%
27%
35%
25%
4%
10%
25%
60%
Predicted

2_generalisation / dg_05_ablation_f1_by_city




Permutation importance by city (drop in test accuracy)
dist_to_center
bedrooms
minimum_nights_avg_ntm
neigh_popularity
accommodates
number_of_reviews
has_dishwasher
bathrooms_qty
score_secure
has_elevator
score_chef_kitchen
has_washing_machine
0.027
0.187
0.042
0.084
0.042
0.024
0.026
0.051
0.035
0.013
0.051
0.043
0.032
0.048
0.036
0.026
0.050
0.023
0.033
0.016
0.035
0.041
0.018
0.024
0.016
0.009
0.018
0.022
0.006
0.023
0.016
0.013
0.008
0.010
0.013
0.011
0.003
0.004
0.013
0.016
0.006
0.010
0.008
0.007
0.004
0.010
0.005
0.010
Paris
Rome
Madrid
Vienna
0.175
0.150
0.125
0.100
0.075
0.050
0.025


Few-shot adaptation to Vienna
3 source cities + n Vienna listings
n Vienna listings only
0 Vienna labels (pure domain generalisation)
F1-macro on held-out Vienna listings (reranked)
0.550
0.525
0.500
0.475
0.450
0.425
0.400
10^2
10^3
Number of labelled Vienna listings used for training (log scale)